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Picture - 1

Normal Equilibrium

↓

Small disturbance (like small load change)

↓

Power System will exhibit oscillations [Nature of oscillation will depend upon loading system, AVR (Pilot value), excitation system]

↓

will oscillations grow or damp out. If oscillation damp out synchronism is preserved and operation is secure, If oscillation grow, power oscillation become severe, protection system may trip the units Instability occurs even w/o fault.

↓

Small signal (analysis) stability (stability of the operating point).

Large system → new steady state PS loaded after disturbance eg. gen load, TL load

Picture 2

PS is operating in steady state:

↓

large disturbance (3ϕ fault) [during the fault electrical power ofp of gen decreases shortly and rotor angle increases and rotor angle increases rapidly transient stability ask questions if the system survives the disturbance

↓

after clearing the fault

↓

If the system survives the disturbance (fault) new operating point is reached otherwise loss of synchronism occurs

↓

Even after the system survives the fault, rotor oscillation continue, generator swing around new operating point. So survival is not enough, post fault system must exhibit stable electromechanical behaviour

↓

Small signal stability of post fault equilibrium point.

Steps Needed [Small Signal Stability Analysis]

1. Linearized model of the system around the operating point
2. Eigen Value analysis.

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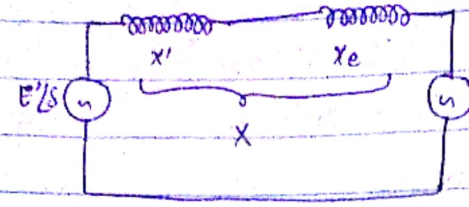
We linearize it around operating point
as at ss - $\delta = \delta_0, \omega = \omega_0, P_m = P_{m0}$
Small disturbance happens → small signal stability - small.

Linearize - we can linearize and approximate it as a linear equation near fixed point (operating point)

Linearized classical SMIB system model

→ all reactance [X]



Swing equation →

$$\frac{d(\delta)}{dt} = \omega - \omega_0 \longrightarrow (1)$$ [actual speed - sync speed]

$$\dot{\delta} = f_1(\omega) \longrightarrow (3)$$ [delta depends upon omega]

$$\frac{d(\omega-\omega_0)}{dt} = \frac{\pi f^0}{H}\left(P_m - \underbrace{\frac{E'E_b \sin\delta}{X}}_{P_e}\right) \longrightarrow (2)$$

$$\dot{\omega} = f_2(P_m, \delta) \longrightarrow (4)$$

Linearized it around operating point →

$$f_1(\omega) = \omega - \omega_0 \qquad f_2(P_m, \delta) = \frac{\pi f^0}{H}\left[P_m - \frac{E'E_b \sin\delta}{X}\right]$$

Linearized model around operating point → [operating point means steady state condition before disturbance]
→ rotor speed constant
→ rotor angle constant

$\delta = \delta^0$
$\omega = \omega^0$ → operating values
$P_m = P_m^0$

$$\Delta\dot{\delta} = \left.\frac{\partial f_1}{\partial \omega}\right|_0 \Delta\omega \rightarrow (5); \quad \Delta\dot{\omega} = \left.\frac{\partial f_2}{\partial P_m}\right|_0 \Delta P_m + \left.\frac{\partial f_2}{\partial \delta}\right|_0 \cdot \Delta\delta \longrightarrow (6)$$

$\Delta y = \left.\frac{\partial f}{\partial x}\right|_0 \Delta x$ for single variable

→ $f_1(\omega) = \omega - \omega_0$ differentiate w.r.t ω

$$\frac{\partial f_1(\omega)}{\partial \omega} = 1 \longrightarrow$$ Substitute this in eq (5)

$\therefore \quad \Delta\dot{\delta} = 1 \cdot \Delta\omega \Rightarrow \boxed{\Delta\dot{\delta} = \Delta\omega} \longrightarrow (7)$

→ $f_2(P_m, \delta) = \frac{\pi f^0}{H}\left(P_m - \frac{E'E_b \sin\delta}{X}\right)$ differential w.r.t P_m

$\frac{\pi f^0}{H}$ and diff w.r.t δ we get →

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$$\frac{\partial P_2}{\partial \delta} = -\frac{\pi f^0}{H} \frac{E' E_b}{X} \cos\delta \qquad \text{[substitute in (6)]}$$

$$\Delta\dot{\omega} = \frac{\pi f^0}{H} \Delta P_m - \frac{\pi f^0}{H} \left[\frac{E' E_b}{X} \cos\delta\right] \longrightarrow (8)$$

$$\Delta\dot{\omega} = \frac{\pi f^0}{H} \Delta P_m - \frac{\pi f^0}{H} K_s \qquad \text{[where } K_s \text{ is synchronizing torque coefficient]}$$

$\therefore$ (9) $\Delta\dot{\delta} = \Delta\omega$, $\Delta\dot{\omega} = \frac{\pi f^0}{H}(\Delta P_m - K_s \Delta\delta)$, (*)

$$\begin{bmatrix} \Delta\dot{\delta} \\ \Delta\dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{\pi f^0 K_s}{H} & 0 \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta\omega \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\pi f^0}{H} \end{bmatrix} \Delta P_m$$

This form depends upon operating point $K_s = \frac{E' E_b}{X} \cos\delta_0$

State : $\dot{X}$ $\quad -A \rightarrow$ (system matrix) $\quad X \quad + \quad B\,u$

[K_s is called synchronizing coefficient, it tells how strongly the generator comes back to synchronism when rotor angle changes slightly. Larger K_s → small angle change produce large corrective electrical power rotor comes back quickly, stability better] Small K_s → weak restoring action, oscillation larger, stability poorer] [$K_s \propto \cos\delta_0$ and δ_0 change with loading, light loading small δ_0, $\cos\delta_0$ large, K_s large good stability ; Heavy loading large δ_0, $\cos\delta_0$ small K_s small poorer stability]

Characteristic equation → $|sI - A| = 0$

$$sI \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} \qquad |sI - A| = \begin{bmatrix} s & -1 \\ \frac{\pi f^0 K_s}{H} & s \end{bmatrix} = 0$$

$$\Rightarrow \left[s^2 + \frac{\pi f^0 K_s}{H} = 0\right] \rightarrow \text{characteristic equation}$$

$s^2 + \omega_n^2 = 0$ $\qquad \omega_n^2 = \frac{\pi f^0 K_s}{H}$ $\qquad \omega_n$ = natural freq of oscillation

$s^2 = -\omega_n^2$

$s^2 = -\frac{\pi f^0 K_s}{H}$

$s = \pm j\sqrt{\frac{\pi f^0 K_s}{H}}$ → purely imaginary, no real part

Oscillation neither grow nor decay they continue forever (sustained oscillation)

Block diagram →

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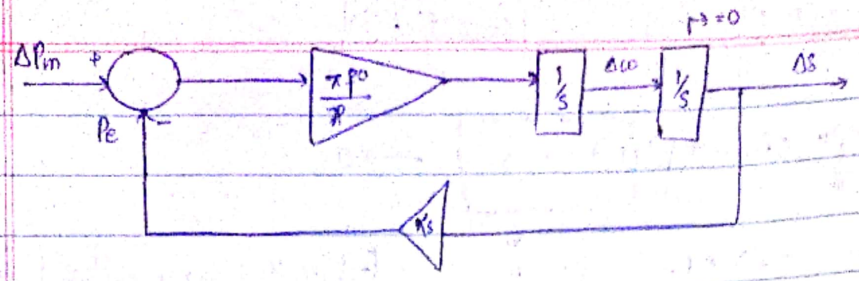
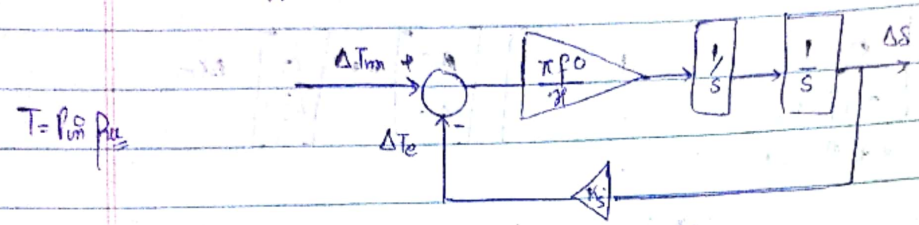


$\Delta\delta = 0$; $\delta = \delta_0$ at operating point (no disturbance yet)

$P_e = \frac{E'E_b}{X} \sin\delta$ $\qquad \Delta P_e = \left(\frac{E'E_b}{X}\cos\delta_0\right)\Delta\delta$



P_m	δ	K_s	Eigen Values	Freq of oscillation	
0.34	20°	0.94	$\pm j\,5.43$	0.864 Hz	(freq of rotor oscillation w.r.t synchronous speed)
0.7	45°	0.71	± 4.69	0.75 Hz	
0.91	60.5°	0.414	$\pm j\,3.6$	0.57	

↑ This shows how loading condition affects small signal stability.

① $\delta_0 \rightarrow$ $\quad P_m = P_e = \frac{E'E_B}{X}\sin\delta_0$ $\quad$ usually $\frac{E'E_B}{X} = 1$

$P_m = P_e = 1 \sin\delta_0$

$0.34 = 1\sin\delta_0$

$\delta_0 = \sin^{-1}(0.34) = 20°$

④ $\rightarrow f = \frac{\omega_n}{2\pi} = \frac{5.43}{2\pi}$ ≈ 0.864 Hz

$\omega_n \rightarrow$ rad/second

② $K_s \longrightarrow \frac{E'E_B}{X}\cos\delta_0$ $\quad \left[\frac{E'E_B}{X} = 1\right]$

$K_s = \cos\delta_0$ $\qquad K_s = \cos(20°)$ ≈ 0.94.

③ Eigen values → $s^2 + \omega_n^2 = 0$

$\omega_n = \pm j\sqrt{\frac{\pi f_0 K_s}{H}}$ $\qquad$ take f_0 as 50 Hz

$H = 5$

$\omega_n \pm j\sqrt{\frac{\pi 50° 0.94}{5}}$

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physically, as soon ... slowly the generator tries to pull the rotor back to synchronous ... angle changes, it's like a restoring force.

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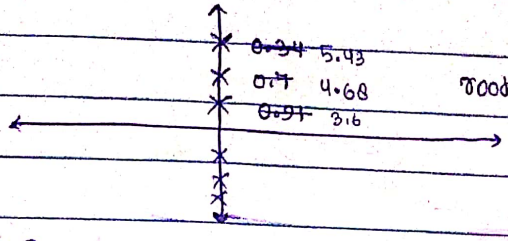
table

As loading Pm ↑ses, δo ↑ses, cos δo decreases, Ks decreases, ωn decreases, freq of oscillation ↓



→ When system is operating at an point – Te (electromagnetic oscillation will be there) — when small disturbance will be there.

→ Behaviour of oscillations depends on system parameters (stiffness decreases)

Practical PS — system → Oscillation will not remain undamped, i.e can grow from oscillation will grow

Relevant Synchronous generator model for small signal stability Analysis.

The classical SMIB linearized model explain the existance of electromechanical oscillations and reveals that their behaviour depends upon the operating point. As the loading increases, the operating rotor angle increases, the synchronising torque coefficient decreases and freq of oscillation also decreases, indicating the weakening of the synchronizing torque with ↑in loading. However, the model assumes constant internal voltage and instantaneous electromagnetic torque response and therefore predicts sustained oscillation.

ΔTe - change in electrical t

ΔTe ← Ks ← Δδ

[Ks constant, this is small signal linearization around operating point]. for small signal Ks constant of disturbance is large, Ks won't stay constant

change in Δδ, change ΔTe we get sustained oscillation

Ks is constant (when disturbance)

In an actual PS, field flux dynamics and AVR effect the electromechanical torque during oscillations causes oscillations to decay, remain poorly damped or even grow. Hence, realistic small signal stability analysis requires a generator mode including the field dynamics and excitation control. The simplest such generator model is 1.0

↳ we can show oscillation decay, poorly damped/even grow.

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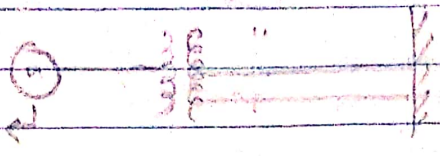
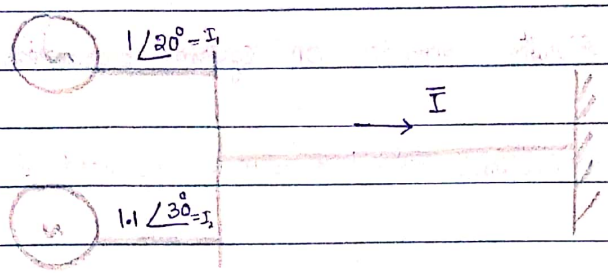
SMIB →



Generator Eb is 1.0

Two important reference frame →

① X-Y reference frame (Synchronous reference frame, Kron's reference frame)



$1\angle 20 \Rightarrow 1(\cos 20 + j \sin 20)$

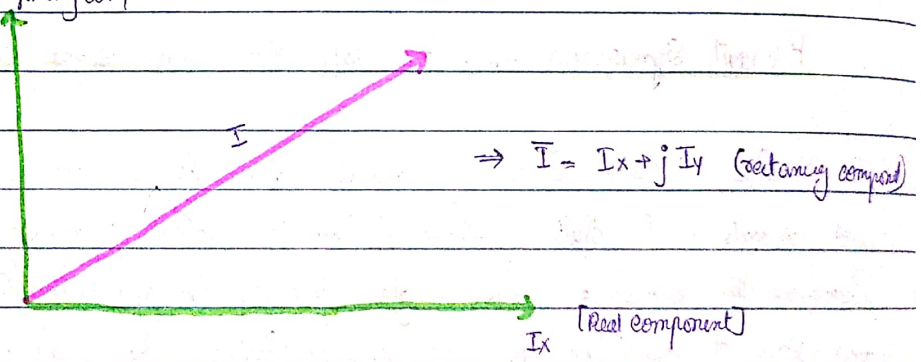
X component, Y component (imaginary component).

$1.1\angle 30° = 1.1\cos 30 + j\,1.1 \sin 30°$

X component, Y component

I_x - cos component

I_y - sin component.



$\Rightarrow \bar{I} = I_x + j I_y$ (rectangular component)

$e^{j\omega_0 t}$ → comes from Eulers formula

$e^{j\omega_0 t} = \cos(\omega_0 t) + j \sin(\omega_0 t)$

This represents a vector rotating in complex plane at angular speed ω_0

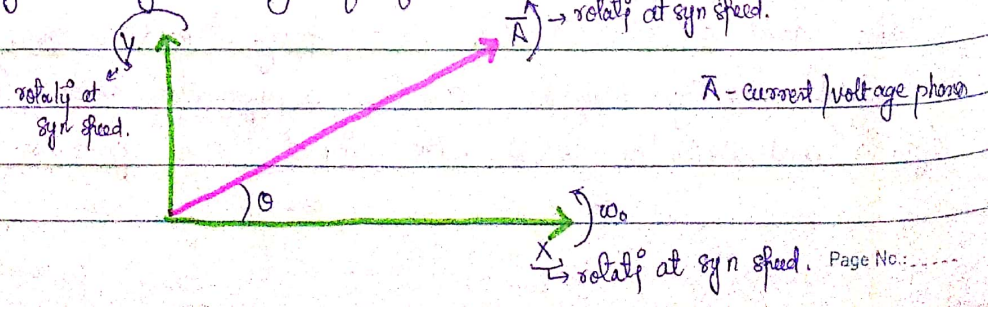
$I_x = 1\cos 20 + 1.1\cos 30° = 0.94 + 0.066 = 1.0923$. (approx).

$I_y = \sin 20 + 1.1 \sin 30° = 0.342 + 0.55 = 0.892$ (approx)

$\bar{I} = 1.0923 + j0.892$

For network/phasor analysis we use X-Y frame

A phasor already assumes removal of $e^{j\omega_0 t}$ [reference frame rotating, this phasor also rotating]. This theory is based on synchronously rotating ref frame



d → direct axis
q → quadrature axis
These axes are attached to the rotor
These axes rotate with rotor

Let $\bar{A}$ is voltage

A voltage phasor $\bar{V} = V_x + jV_y$

$V_x = V\cos\theta$ $\quad V_y = V\sin\theta$

$V_x + jV_y = V\cos\theta + V\sin\theta \Rightarrow V\angle\theta$

→ voltage/current that look sinusoidal in stationary frame become nearly constant (DC-like).

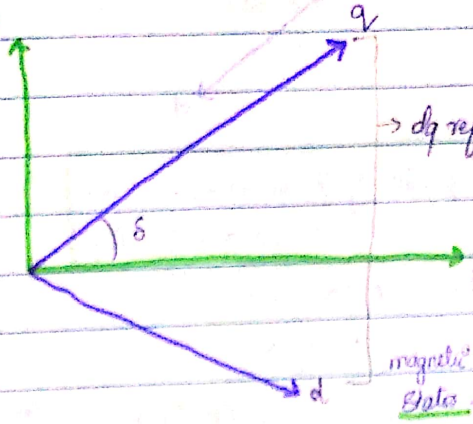
Let $\bar{A}$ is current →

Similarly → $\bar{I} = I_x + jI_y$

⊙ d-q reference frame (rotor reference frame / Parks reference frame).

rotate with generator rotor, steady state - synchronous speed. Generally → no-syn speed.
as they are attached to the rotor rotor speed = synchronous speed ↳ during disturbance

Standard synchronous speed model (like 1.0) derived or available for d-q reference frame (as model becomes much simple)

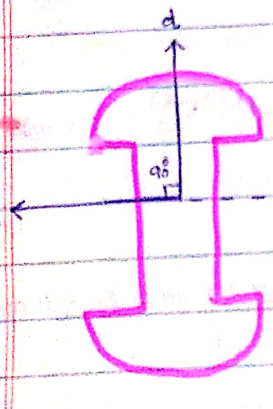


δ - angle b/w q and x.

→ dq reference frame

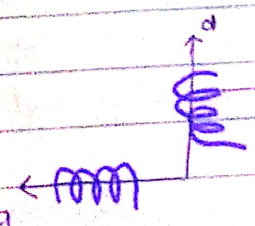
Rotor - rotor carries DC excitation → produces magnetic field. As rotor physically rotates, thus field rotates. So rotor magnetic field rotates mechanically

Stator - stator has 3 phase windings → phase a, b, c shifted 120°



q is leading d by 90° [rotor stator field is rotating & stator winding are shifted by 120° (voltage & current)]

→ Why d axis and q axis coils? as 3 phase stator currents are difficult to a, b, c all sinusoidal and changing. So Park said → convert them to 2 equivalent windings → d axis winding, q axis winding
→ these coils produce same field as stator field



stator 2 coil (one on d, other q) coil produce same rotating field as stator, ~~same field~~.

(coil) ← these are rotating → produce same field as the field produced by stator

a phase, b phase and c phase - combined rotating magnetic field = two imaginary coils could do same net magnetic effect

Voltage - V_d, V_q → $V_d + jV_q$
Current - I_d, I_q → $I_d + jI_q$ → wrong (we can not represent them like this d = real part, q = imaginary part)

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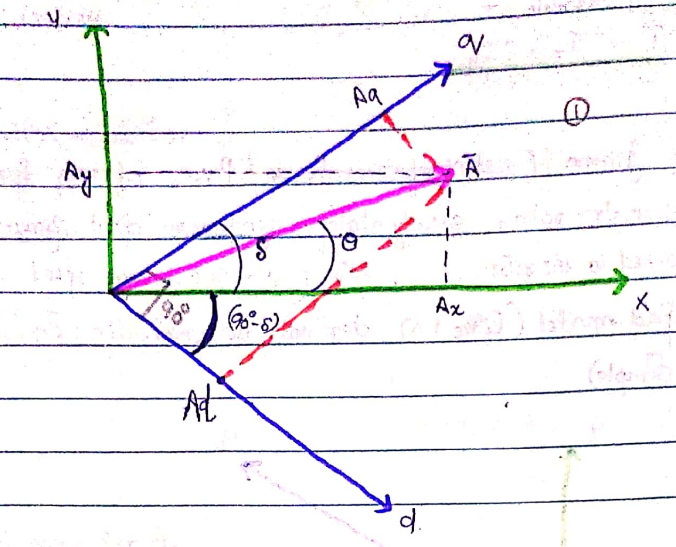
$\bar{V} = V_x + j V_y$ ✓ → (True for XY plane)

$V \neq V_d + j V_q$ ✗ (we can't write $\bar{V}$ in this form, we have to transform it to dq to XY ref frame)

Relation of XY with DQ dq →



q - δ (q is leading)

$\bar{A}$ → means current/voltage

$\theta = \bar{A}$ and x (+ve)

$AX - d = (90 - \delta)$

Ax add diff of $(90-\delta)$

$\bar{A} = A_x + A_y$. ✓

$\bar{A} = A_d + j A_q$ ✗ (wrong) as it's lagging by $(90-\delta)$

Projection of A on d-axes = Ad ②

" " " " q axis = Aq

$\bar{A}x + j AY = A\angle\theta$ ✓

$\bar{A}d + j Aq \neq A\angle\theta$ ✗ ③

$Ad + jAq = A\angle\theta - (90-\delta)$ → $Ad + jAq = A\angle\theta\, e^{-j(90-\delta)}$

$A = (Ad + j Aq)\, e^{j(\pi/2 - \delta)}$ (advances) ★

$Ae^{\pm j\theta} = \bar{A} = AX + j AY$ ★

$AX + jAY = (Ad + jAq)\, e^{j(\frac{\pi}{2} - \delta)}$ we know, A make θ with X ; D makes $(90^\circ - \delta)$ with X. ∴ relative angle b/w A and D = $[\theta + (90-\delta)]$

$Ad = A\cos(90^\circ - \delta + \theta)$. [vector projection].

$Aq = A \sin(90^\circ - \delta + \theta)$

Adding → $Adq = Ad + jAq = A\angle\theta\,/\,90-\delta\; e^{j(\delta - \pi/2)}$

(5) $\dfrac{(Id + jIq)\, e^{j(\delta - \pi/2)}}{Ix + j\, Iy}$

(5) $\dfrac{(Vd + jVq)\, e^{-j(\delta - \pi/2)}}{Vx + j\, Vy}$

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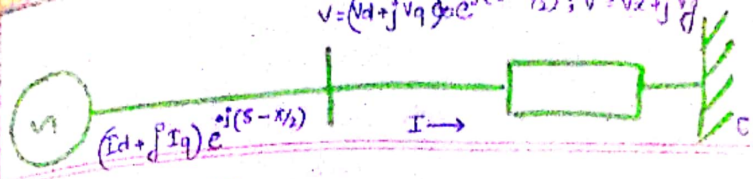


$\bar{I} = Ix + jIy$

Applying KVL = $V = Eb + jXI$

$(Vd+jVq)e^{j(\delta-\pi/2)} = Eb\angle 0 + jX(Id+jIq)e^{j(\delta-\pi/2)}$

Notation → $e^{j\theta} = \cos\theta + j\sin\theta$; $e^{j(\delta-\pi/2)} = \cos(\delta-\frac{\pi}{2}) + j\sin(\delta-\pi/2)$

$\cos(\delta-\frac{\pi}{2}) = \sin\delta$; $\sin(\delta-\frac{\pi}{2}) = -\cos\delta$

$e^{j(\delta-\pi/2)} = \sin\delta - j\cos\delta$

$\therefore I = (Id+jIq)(\sin\delta - j\cos\delta) \Rightarrow Id\sin\delta - jId\cos\delta + jIq\sin\delta - (j)^2 Iq\cos\delta$

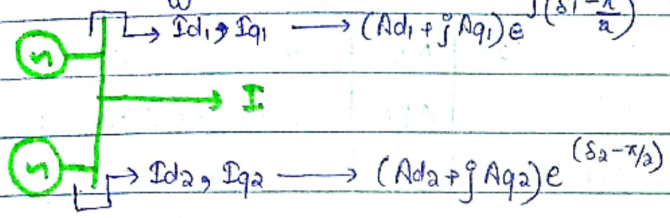
$Id\sin\delta - jId\cos\delta + jIq\sin\delta - (-1)^2 Iq\cos\delta$

$Id\sin\delta - Id\cos\delta + jIq\sin\delta + Iq\cos\delta$

$Id\sin\delta + jIq\sin\delta.$

$$\begin{bmatrix} Vx \\ Vy \end{bmatrix} = \underbrace{\begin{bmatrix} \sin\delta & \cos\delta \\ -\cos\delta & \sin\delta \end{bmatrix}}_{T(\delta)} \begin{bmatrix} Vd \\ Vq \end{bmatrix} ; \quad \begin{bmatrix} Vd \\ Vq \end{bmatrix} = \begin{bmatrix} \sin\delta & -\cos\delta \\ \cos\delta & \sin\delta \end{bmatrix} \begin{bmatrix} Vx \\ Vy \end{bmatrix}$$

Diff generators have different (d-q) frame →



$Vq = Vx\cos\delta + \cdots$

$Vd = Vx\sin\delta - Vy\cos\delta$

$Id = I_2$

Q Given → $\bar{I} = 1.0526\angle -18.80°$; $\bar{V} = 1.0946\angle 11.59°$; $\delta = 52.1°$

find $Id, Vd, Vq, Ix, Iy, Vx, Vy$ ~~Vt~~ Iq

Sol → $Ix = 1.0526\cos(-18.80°) = 0.996$

$Iy = 1.0526\sin(-18.80°) = -0.340$

$\therefore I = 0.996 - j0.340$

$Vx = 1.0946\cos(11.59°) = 1.072$

$Vy = 1.0946\sin(11.59°) = 0.220$

$\therefore V = 1.072 + j0.220.$

$Vd = Id = Ix \sin\delta + Iy \cos\delta = 0.996\cos 52.1° + (-0.340)\cos(52.1)° =$

$Iq = 0.996$

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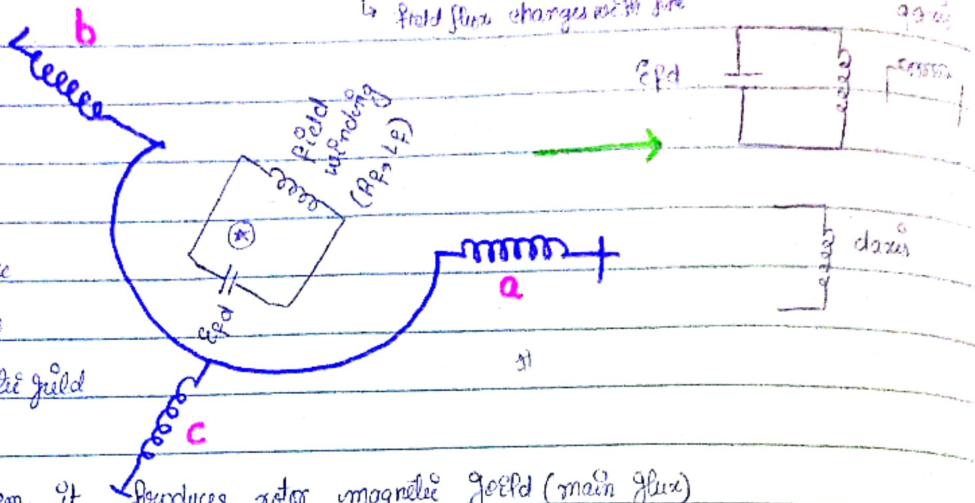
Small signal model of SMIB system

SMIB synchronous machine – flux decay model (Model 1.0)

↳ field flux changes with time



* DC excitation system, it produces rotor magnetic field (main flux)

Rotor has field winding (DC excitation), DQ reference frame is attached to rotor, D and Q-axis rotate together with rotor field ∴ DQ frame rotates synchronously with rotor field.

We consider → no damping winding.

Linearized model for classical model

$\dot{X} = Ax + bu$

rotor angle deviation	$x =$	$\Delta\delta$	$u =$	ΔT_m	→ mechanical I/P
speed deviation	←	$\Delta\omega$		ΔE_{fd}	→ excitation system I/P (field excitation voltage)
flux (EMF) deviation	←	$\Delta E'_q$			ΔE_{fd} – controls field flux (voltage regulator).

$\Delta E'_q$ – flux dynamics

rotor internal EMF is not constant

→ It changes due to excitation + armature reaction

∴ E_q' is not constant (internal emf)

→ E_{fd} → field excitation voltage —(controls)→ field current → flux → E' → V_t (terminal voltage)

Terminal voltage – voltage at generator O/P.

Case ① → E_{fd} (constant) → excitation is fixed, rotor flux is fixed, internal emf E' is fixed, so when load changes V_t will change. Result → [no control, poor voltage regulation].

Case ② → E_{fd} changes (AVR ON) → AVR continuously measures V_t $(V_{ref} - V_t)$

Control action → V_t drops → AVR ↑ses E_{fd}.

V_t rises → AVR ↓ses E_{fd}.

E_{fd} directly controls flux

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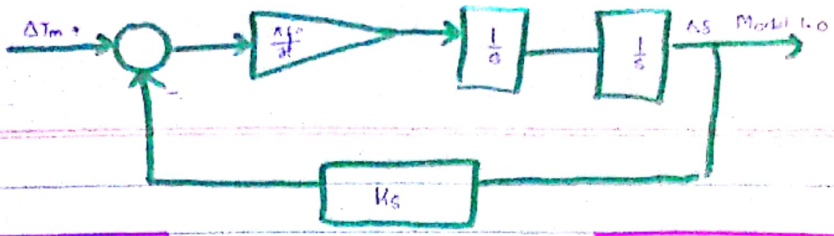


$\dot{\delta} = \omega - \omega_0$ → (4)

$\dot{\omega} = \frac{\pi f_0}{H} (P_m - P_e)$ → (5)

$S = \bar{V}(\bar{I}^*) \rightarrow P = Re(S) \rightarrow S = [V_x + jV_y][I_x - jI_y] \rightarrow P = Re(S) = V_x I_x + V_y I_y$

$P = Re(S) = V_d I_d + V_q I_q$

From above diagram → $\bar{V} = V_x + jV_y$; $\bar{V} = (V_d + jV_q) e^{j(\delta - \pi/2)}$

$\bar{I} = I_x + jI_y$; $\bar{I} = (I_d - jI_q) e^{j(\delta - \pi/2)}$

$S = (V_d + jV_q) e^{j(\delta - \pi/2)} \times (I_d - jI_q) e^{-j(\delta - \pi/2)} \rightarrow I^*$

$S = (V_d + jV_q)(I_d - jI_q)$

$S = V_d I_d - jV_d I_d + jV_q I_q - (j)^2 V_q I_q$

$P = V_d I_d + V_q I_q$ → (3)

V_d, V_q → terminal voltage component on DQ

Stator equation

Stator voltage equations in DQ frame →

$V_d = X_q I_q$ → (1)

↓ q-axis current affects d-axis voltage

$V_q = E_q' - X_d' I_d$ → (2)

↓ d-axis current effects q axis voltage

E_q' change due field excitation system (E_{fd}), armature rxn, rotor flux dynamics

stronger flux — higher E_q' | weak flux — lower E_q'

Excitation ↑ → E_q', armature rxn ↓ → E_q' ↑,

Field equation →

E_{fd} (excitation) → $\frac{1}{1 + sT_{do}'}$ → E_q' (flux)

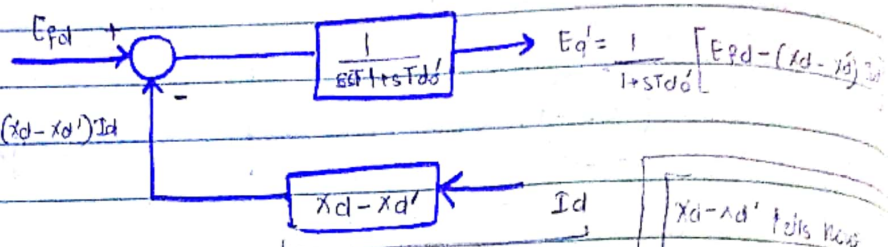
stator - no current - no armature rxn

Physical meaning - field winding cannot change instantly, flux builds slowly, flux decays slowly

→ Current flowing in the stator — armature ran.

Stator current produces flux that opposes rotor field flux, reduces effective excitation so its modeled as $(X_d - X_d')I_d$.

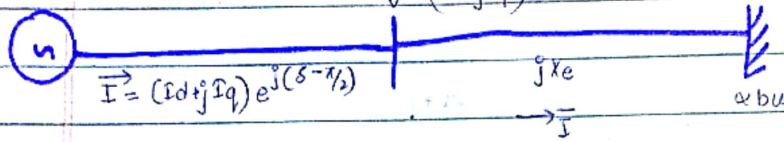
Combined model →



$X_d - X_d'$ is less as much reactance changes upon stator

$(X_d - X_d')I_d$ - voltage term caused by d-axis I when Id changes Emf chan

Network dynamics / Model →



Apply KVL →

$$(V_d + jV_q)e^{j(\delta-\pi/2)} = jX_e(I_d + jI_q)e^{j(\delta-\pi/2)} + E_b$$

multiply b/s by $e^{-j(\delta-\pi/2)}$

$$V_d + jV_q = jX_e(I_d + jI_q) + E_b e^{-j(\delta-\pi/2)}$$

$$(V_d + jV_q) = jX_e I_d - X_e I_q + E_b \cos e^{-j(\delta-\pi/2)}$$

$E_b e^{-j(\delta-\pi/2)}$

$e^{-j\theta} = \cos\theta - j\sin\theta$

$E_b[\cos(\delta - \frac{\pi}{2}) - j\sin(\delta - \frac{\pi}{2})]$

$E_b[\cos(\delta - \frac{\pi}{2})$

$E_b[\sin\delta - j\cos(-\cos\delta)]$

$E_b[\sin\delta + j\cos\delta]$

$$(V_d + jV_q) = jX_e I_d - X_e I_q + E_b \sin\delta + jE_b\cos\delta$$

equate real and imaginary →

(6) → $V_d = E_b \sin\delta - X_e' I_q$ → Real part

(7) → $V_q = E_b\cos\delta + X_e' I_d$ → Imaginary part.

∴ Using (7) in (2) we get →

$$E_q' - X_d' I_d = E_b\cos\delta + X_e I_d$$

$$-X_e I_d - X_d' I_d = E_b\cos\delta - E_q'$$

$$(-X_d' - X_e) I_d = E_b\cos\delta - E_q'$$

Step (1) →

$$I_d = \frac{E_q' - E_b\cos\delta}{X_d' + X_e}$$

constant

→ (8)

$f_1(\delta, E_q')$

→ diff

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Using (1) and (6)

$$E_b \sin\delta - x_e I_q = x_q I_q$$

$$E_b \sin\delta = x_q I_q + x_e I_q$$

$$E_b \sin\delta = (x_q + x_e) I_q$$

Step (2) →

$$\boxed{\frac{E_b \sin\delta}{x_q + x_e} = I_q} \longrightarrow (9) \qquad f_2(\delta).$$

$$\Delta \dot{\psi}_d = \left.\frac{\partial f_1}{\partial \delta}\right|_0 \Delta\delta + \left.\frac{\partial f_1}{\partial E_q'}\right|_0 \Delta E_q'$$

$$\frac{\partial}{\partial \delta}\left(\frac{E_q'}{x_d' + x_e}\right) = -\frac{E_b \cos\delta}{x_d' + x_e}$$

$$= 0 - \frac{E_b(-\sin\delta)}{x_d' + x_e} = \frac{E_b \sin\delta}{x_d' + x_e}$$

$$\Delta \dot{I}_d = \underbrace{\left.\frac{E_b \sin\delta_0}{x_e + x_d'}\right|_0}_{c1} \Delta\delta + \underbrace{\left.\frac{1}{x_e + x_d'}\right|_0}_{c2} \Delta E_q'$$

Step 3 :

$$\boxed{\Delta \dot{I}_d = c_1 \Delta\delta + c_2 \Delta E_q'} \longrightarrow (10)$$

$$\Delta \dot{I}_q = \left.\frac{\partial f_2}{\partial \delta}\right|_0 \Delta\delta \Rightarrow \left.\frac{E_b \cos\delta_0}{x_e + x_q}\right|_0 \Delta\delta$$

c_3

Step (4) →

$$\boxed{\Delta \dot{I}_q = c_3 \Delta\delta} \longrightarrow (11)$$

Electrical torque of the generator T_e (no i)

In DS model $T_e = V_d I_d + V_q I_q$ [we got this from eq (3)]

$T_e = x_q I_q I_d + [E_q' - x_d' I_d] I_q$ → [from eq (1) and (2)]

$T_e = x_q I_q I_d + E_q' I_q - x_d' I_d I_q$

$$\boxed{T_e = (x_q - x_d') I_q I_d + E_q' I_q} \longrightarrow (12)$$

$T_e = f_3(E_q', I_q, I_d)$ (q; att, abs, 4, 4, 8)

Step (5) →

$$\Delta T_e = \left.\frac{\partial f_3}{\partial E_q'}\right|_0 \Delta E_q' + \left.\frac{\partial f_3}{\partial I_d}\right|_0 \Delta I_d + \left.\frac{\partial f_3}{\partial I_q}\right|_0 \Delta I_q$$

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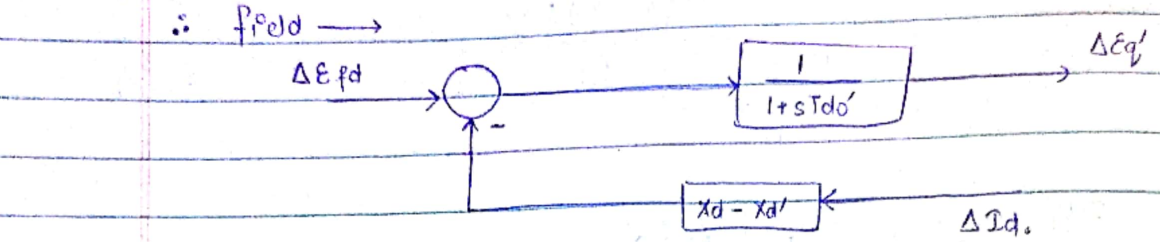
$$\Delta T_e = I_{qo}\,\Delta E_q' + \left[(X_q - X_d')\,I_{qo}\right]\Delta I_d + \left[E_q'^{o} + (X_q - X_d')\,I_{do}\right]\Delta I_q$$

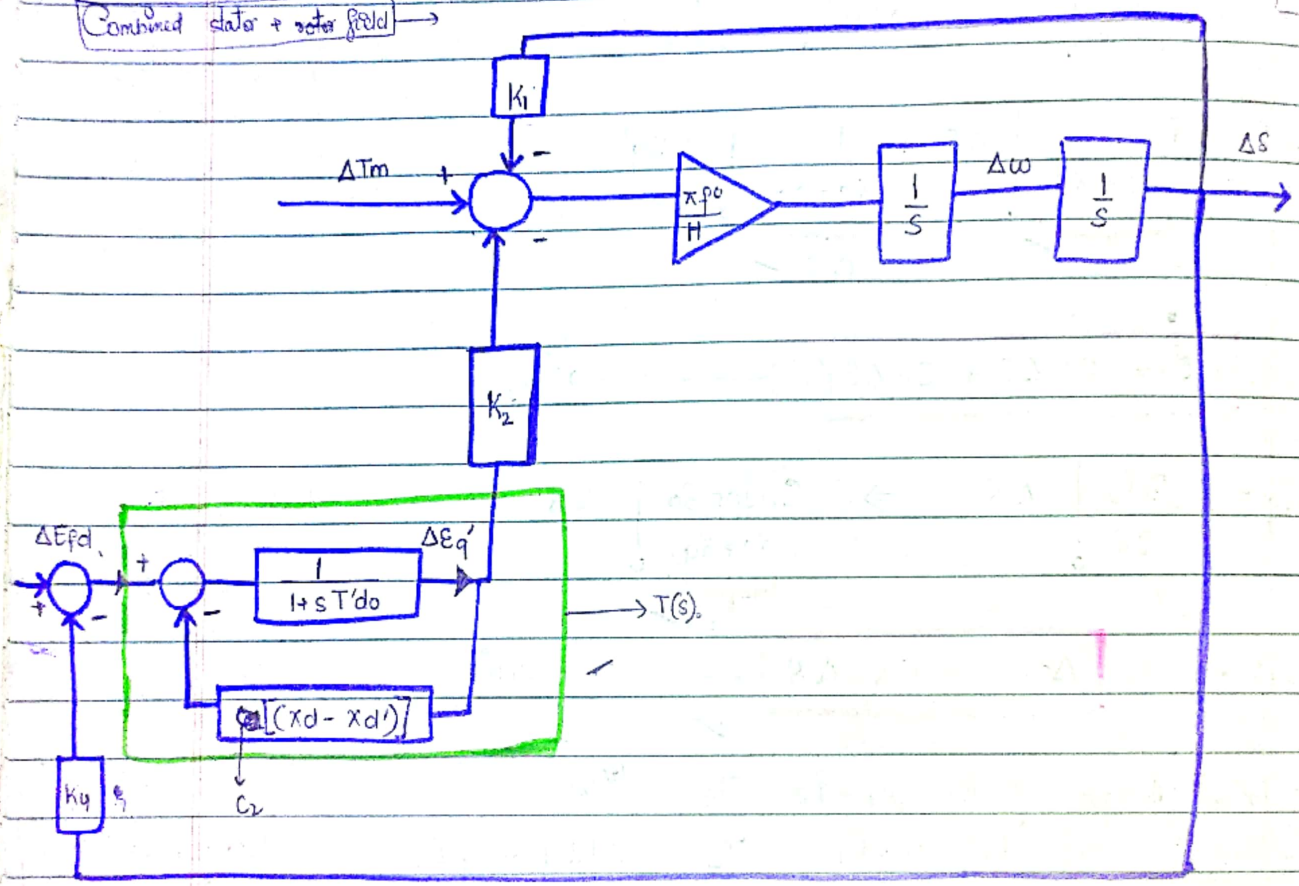
solve.

$$\Delta T_e = K_1 \Delta\delta + K_2 \Delta E_q' \longrightarrow (B)$$

$\therefore$ field $\longrightarrow$



Combined data + rotor field $\longrightarrow$



$$T_s = \frac{\dfrac{1}{1+sT'_{do}}}{1 + \dfrac{C_2 (X_d - X_d')}{1+sT'_{do}}}$$

$$T_s \Rightarrow \frac{\dfrac{1}{1+sT'_{do}}}{\dfrac{1+sT'_{do} + C_2 (X_d - X_d')}{1+sT'_{do}}}$$

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$T_s \Rightarrow \dfrac{1}{1 + sT'd_0 + C_2(X_d - X_d')}$ $\qquad T_s = \dfrac{1}{[1 + C_2(X_d - X_d')] + sT'd_0}$

devide numerator and denominator by $1 + C_2(X_d - X_d')$

$$T_s = \frac{\boxed{\dfrac{1}{1 + C_2(X_d - X_d')}} \Rightarrow K_3}{\dfrac{1 + sT'd_0}{(1 + C_2(X_d - X_d'))}} \rightarrow \boxed{T_s = \frac{K_3}{1 + sT'd_0 K_3}}$$

$\boxed{K_3 = \dfrac{1}{1 + C_2(X_d - X_d')}}$ $\qquad \boxed{K_4 = (X_d - X_d')C_1}$

$\Delta E_q'$

ΔS

Assignment / Homework. →

$$\Delta T_e = I_{q0}\Delta E_q' + (X_q - X_d')I_{q0}\Delta I_d + E_q'{}_0\,\Delta I_q + (X_q - X_d')I_{d0}\Delta I_q$$

$$\Delta T_e = I_{q0}\Delta E_q' + (X_q - X_d')I_{q0}\Delta I_d + \left[E'_{q0} + (X_q - X_d')I_{d0}\right]\Delta I_q$$

from eq (10) – $\Delta I_d = C_1\Delta S + C_2\Delta E_q'$ $\qquad$ from eq (11) → $\Delta I_q = C_3\Delta S$

substituting above

$$\Delta T_e = I_{q0}\Delta E_q' + (X_q - X_d')I_{q0}(C_1\Delta S + C_2\Delta E_q') + \left[E'_{q0} + (X_q - X'_d)I_{d0}\right]C_3\Delta S$$

$$\Delta T_e = I_{q0}\Delta E_q' + (X_q - X_d')I_{q0}C_1\Delta S + (X_q - X_d')I_{q0}C_2\Delta E_q' + \left[E'_{q0} + (X_q - X'_d)I_{d0}\right]C_3\Delta S$$

$$\Delta T_e = I_{q0}\Delta E_q' + \Delta S\left[(X_q - X_d')I_{q0}C_1 + \left(E'_{q0} + (X_q - X'_d)I_{d0}\right)C_3\right] + (X_q - X_d')I_{q0}C_2\Delta E_q'$$

$\Delta T_e \rightarrow$ $\boxed{K_1 = \left[(X_q - X_d')I_{q0}C_1 + \left(E'_{q0} + (X_q - X'_d)I_{d0}\right)C_3\right]}$

$$\Delta T_e = K_1\Delta S + I_{q0}\Delta E_q' + (X_q - X_d')I_{q0}C_2\Delta E_q'$$

$$\Delta T_e = K_1\Delta S + \Delta E_q'\left[I_{q0} + (X_q - X_d')I_{q0}C_2\right]$$

$\boxed{K_2 = \left[I_{q0} + (X_q - X_d')I_{q0}C_2\right]}$

$$\Delta T_e = K_1\Delta S + K_2\Delta E_q'$$

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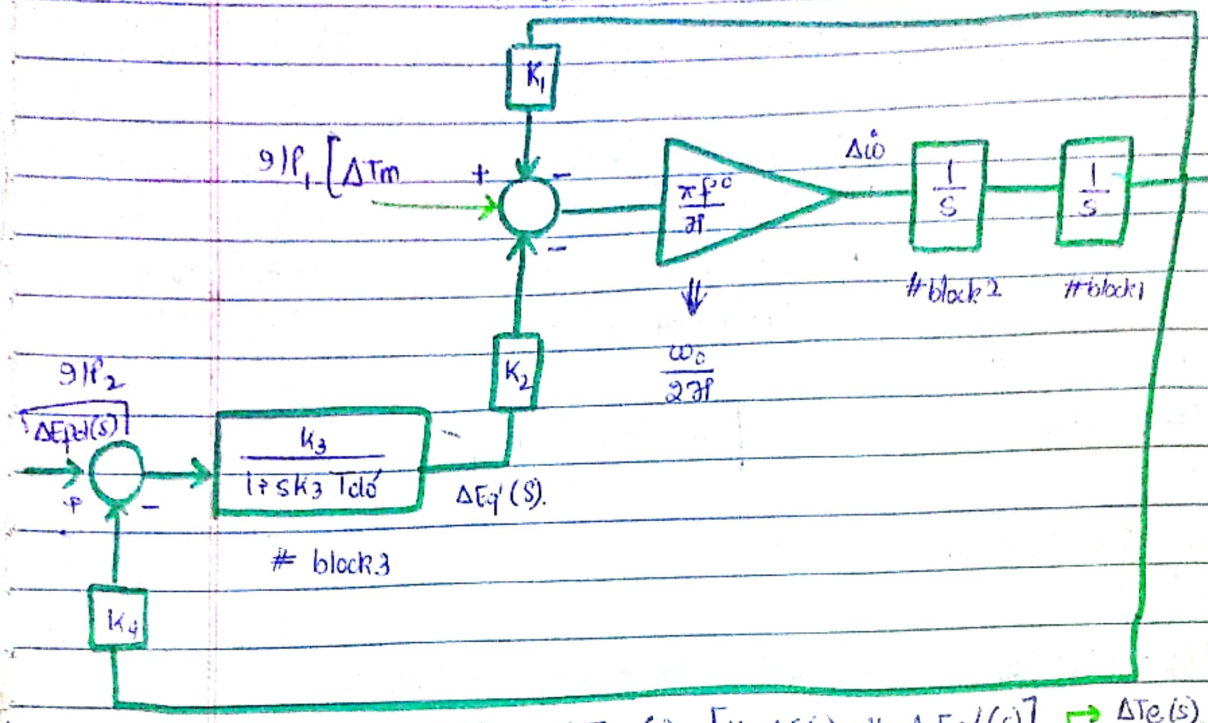
Small Signal model for SMIB [with flux decay model - model in block diagram form] without AVR.



Here, $\Delta T_m(s) - \Delta T_e(s) = \Delta T_m(s) - [K_1 \Delta S(s) + K_2 \Delta E_q'(s)] \rightarrow \Delta T_e(s)$.

$\dot{X} = AX + B(u)$ → then we find eigen value of (A) to check weather the system is stable or not. A matrix will be 3x3.

State variable representation →

Block 1 → $\Delta \dot{S} = \Delta \omega$ → ①

Block 2 → $\Delta \dot{\omega} = \frac{\omega_0}{2H} (\Delta T_m - K_1 \Delta S - K_2 \Delta E_q')$ → ②

$\frac{\text{Sff}}{\text{Sff}} = \text{Block}$

Block 3 → $\frac{\Delta E_q'}{\Delta E_{fd} - K_4 \Delta S} = \frac{K_3}{1 + K_3 T_{do}' \frac{d}{dt}}$ (time domain)

$\Delta E_q + \Delta \dot{E}_q K_3 T_{do}' = K_3 \Delta E_{fd} - K_3 K_4 \Delta S$

$\Delta \dot{E}_q = \frac{K_3 \Delta E_{fd}}{K_3 T_{do}'} - \frac{K_3 K_4 \Delta S}{K_3 T_{do}'} - \frac{\Delta E_q}{K_3 T_{do}'}$

$\Delta \dot{E}_q' = \frac{\Delta E_{fd}}{T_{do}'} - \frac{K_4 \Delta S}{T_{do}'} - \frac{\Delta E_q}{K_3 T_{do}'}$ → ③

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$$\begin{bmatrix}\Delta\dot{\delta}\\ \Delta\dot{\omega}\\ \Delta\dot{E}_q'\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ \frac{-K_1\omega_0}{2H} & 0 & \frac{-K_2\omega_0}{2H}\\ \frac{-K_4}{T_{do}'} & 0 & \frac{-1}{K_3 T_{do}'}\end{bmatrix}\begin{bmatrix}\Delta\delta\\ \Delta\omega\\ \Delta E_q'\end{bmatrix} + \begin{bmatrix}0 & 0\\ \frac{\omega_0}{2H} & 0\\ 0 & \frac{1}{T_{do}'}\end{bmatrix}\begin{bmatrix}\Delta T_m\\ \Delta E_{fd}\end{bmatrix}$$

↓

state matrix form for small signal [without AVR]

Effect of excitation system →

[reference] V_{ref} → ⊕ (− V) → $\frac{K_e}{1+sT_e}$ → E_{fd} → (voltage across field winding)

V [actual terminal voltage of generator]

→ We will now extend the block diagram developed for SMIB [with model 1.0] to include the effects of excitation system (AVR).

ΔV_{ref} + → ⊕ (− ΔV) → $\frac{K_e}{1+sT_e}$ → ΔE_{fd}

T_e [Same for (xy) (DQ) plane] $= V_x I_x + I_y V_y = V_d I_d + V_q I_q$

XY DQ

Value of terminal voltage

$V = \sqrt{V_x^2 + V_y^2}$

$V = \sqrt{V_d^2 + V_q^2}$

$\rightarrow V = f(V_d, V_q)$ ⇒

$\therefore \Delta V = K_5 \Delta\delta + K_6 \Delta E_q'$

∴ with AVR we have 6 constant $(K_1, K_2, K_3, K_4, K_5, K_6)$ ↦ Heffron Phillip constant.

$$\Delta V = \left.\frac{\partial f}{\partial V_d}\right|_0 \Delta V_d + \left.\frac{\partial f}{\partial V_q}\right|_0 \Delta V_q$$

$$\Delta V = \left.\frac{1}{2\sqrt{V_{d0}^2+V_{q0}^2}} \times 2V_d\right|_0 \Delta V_d + \left.\frac{1}{2\sqrt{V_{d0}^2+V_{q0}^2}} \times 2V_q\right|_0 \Delta V_q$$

$$\Delta V = \frac{V_{do}}{\sqrt{V_{do}^2+V_{q0}^2}}\Delta V_d + \frac{V_{qo}}{\sqrt{V_{d0}^2+V_{q0}^2}}\Delta V_q \Rightarrow \Delta V = \frac{V_{do}}{V_o}\Delta V_d + \frac{V_{qo}}{V_o}\Delta V_q \quad \left[V_o = \sqrt{V_{d0}^2+V_{q0}^2}\right]$$

$$\rightarrow \Delta V = \frac{V_{do}}{V_o}\Delta V_d + \frac{V_{qo}}{V_o}\Delta V_q$$

[But we want in the form of $\Delta V = K_5\Delta\delta + K_6\Delta E_q'$]

We know $V_d = X_q I_q$ and $\Delta V_q = \Delta E_q' - X_d' I_d$.

$= V_d = X_q C_3 \Delta\delta$ $\Delta V_q = \Delta E_q' - X_d' [C_4\Delta\delta + C_2\Delta E_q']$

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$$\Delta V = \frac{V_{do}}{V_o}\left[x_q C_3 \Delta\delta\right] + \frac{V_{qo}}{V_o}\left[\Delta E_q' - x_d'\left(C_1\Delta\delta + C_2\Delta E_q'\right)\right]$$

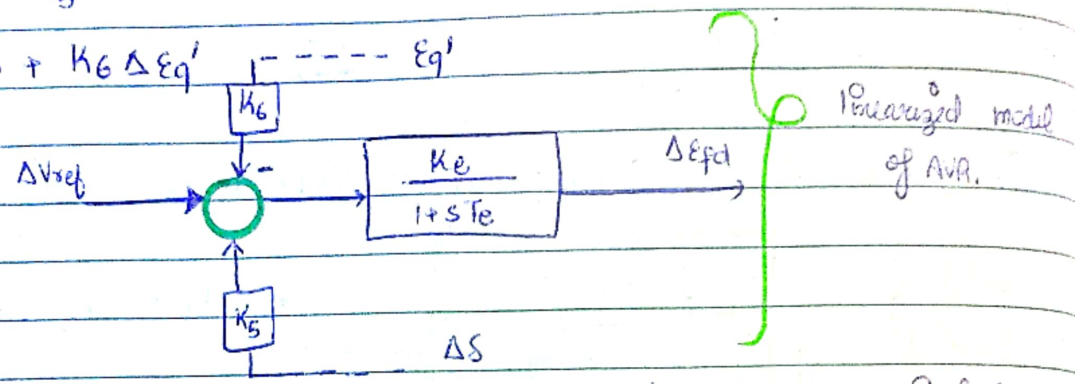
$$\Delta V = \frac{V_{do}}{V_o}\left[x_q C_3 \Delta\delta\right] + \frac{V_{qo}}{V_o}\left[\Delta E_q' - C_1\Delta\delta x_d' - C_2\Delta E_q' x_d'\right]$$

$$\Delta V = \frac{V_{do}}{V_o}\left(x_q C_3 \Delta\delta\right] + \Delta E_q'\frac{V_{qo}}{V_o} - \frac{V_{qo}}{V_o}C_1\Delta\delta x_d' - \frac{V_{qo}}{V_o}C_2\Delta E_q' x_d'$$

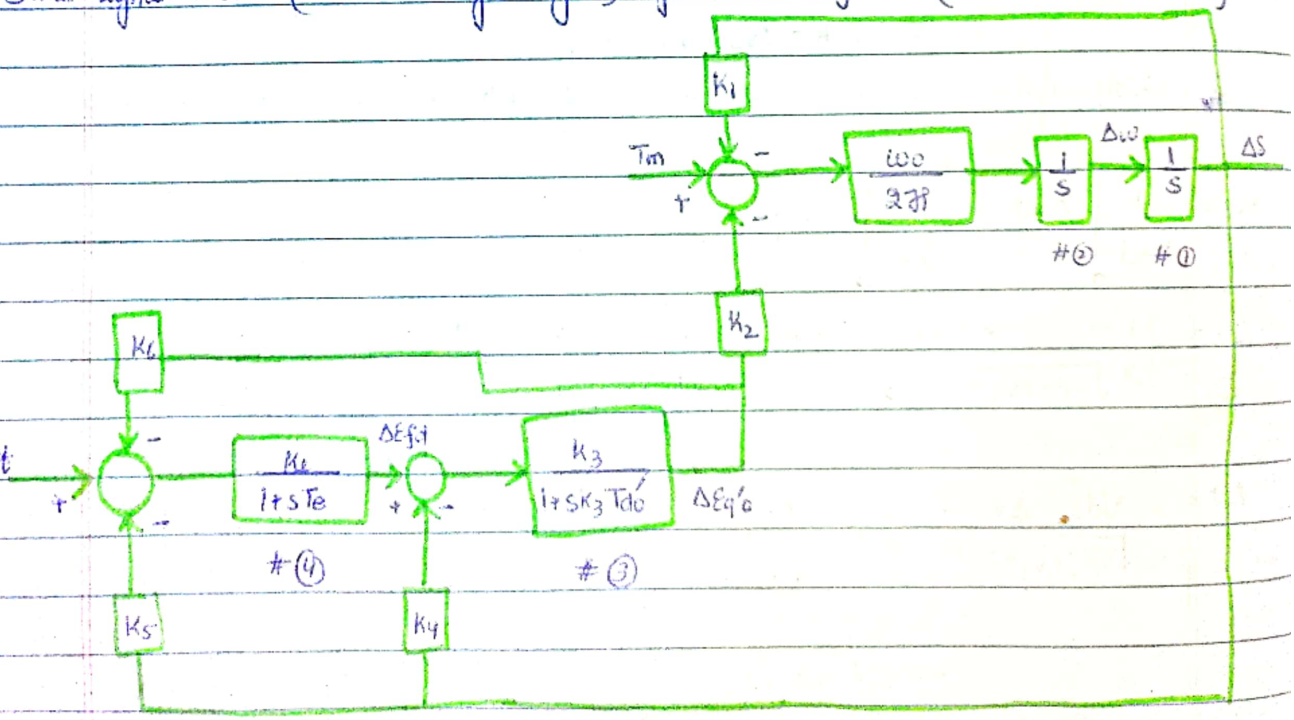
$$\left[\frac{V_{do}}{V_o}x_q C_3 - \frac{V_{qo}}{V_o}C_1 x_d'\right]\Delta\delta + \left[\frac{V_{qo}}{V_o} - \frac{V_{qo}}{V_o}C_2 x_d'\right]\Delta E_q'$$

$$\Delta V = \underbrace{\left[\frac{V_{do}}{V_o}x_q C_3 - \frac{V_{qo}}{V_o}C_1 x_d'\right]}_{K_5}\Delta\delta + \underbrace{\left[\frac{V_{qo}}{V_o}\left[1 - C_2 x_d'\right]\right]}_{K_6}\Delta E_q'$$

$$\Delta V = K_5\Delta\delta + K_6\Delta E_q' \quad \text{---- } E_q'$$



→ Small signal model (block diagram form) for SMIB system (model 1.0) including AVR



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we check stability for different loading change [Pm = 0.5, Pm = 1.0 ...]
eigen value may diff for [Pm = 0.5, Pm = 1.0].

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Matrix form →

#1 $\Delta\dot{\delta} = \Delta\omega$ ——→ (1)

#2 $\Delta\dot{\omega} = \frac{\omega_0}{2H}\left(\Delta T_m - K_1\Delta\delta - K_2\Delta E_q'\right)$ ——→ (2)

#3 $\Delta\dot{E}_q' = \frac{-E_q'}{K_3 T_{do}'} + \frac{\Delta E_{fd}}{T_{do}'} - \frac{K_4\Delta\delta}{T_{do}'}$ ——→ (3)

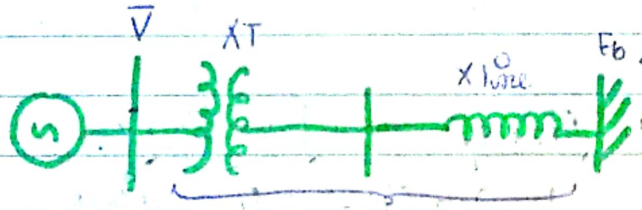
#4 $\frac{\Delta E_{fd}}{\Delta V_{ref} - K_6\Delta E_q' - K_5\Delta\delta} = \frac{K_e}{1 + T_{ed}\frac{d}{dt}}$

$\rightarrow \Delta\dot{E}_{fd} = \frac{\Delta E_{fd}}{T_e} + \frac{K_e\Delta V_{ref}}{T_e} - \frac{K_5 K_e\Delta\delta}{T_e} - \frac{K_6 K_e\Delta E_q'}{T_e}$ ——→ (4)

$$\begin{bmatrix}\Delta\dot{\delta}\\ \Delta\dot{\omega}\\ \Delta\dot{E}_q'\\ \Delta\dot{E}_{fd}\end{bmatrix} = \underbrace{\begin{bmatrix}0 & 1 & 0 & 0\\ \frac{-K_1\omega_0}{2H} & 0 & \frac{-K_2\omega_0}{2H} & 0\\ \frac{-K_4}{T_{do}'} & 0 & \frac{-1}{K_3 T_{do}'} & \frac{1}{T_{do}'}\\ \frac{-K_5 K_e}{T_e} & 0 & \frac{-K_6 K_e}{T_e} & \frac{-1}{T_e}\end{bmatrix}}_{A} \begin{bmatrix}\Delta\delta\\ \Delta\omega\\ \Delta E_q'\\ \Delta E_{fd}\end{bmatrix} + \begin{bmatrix}0 & 0\\ \frac{\omega_0}{2H} & 0\\ 0 & 0\\ 0 & \frac{K_e}{T_e}\end{bmatrix}\begin{bmatrix}\Delta T_m\\ \Delta V_{ref}\end{bmatrix}$$

A → to check stability (4 × 4).

(2 machine) →



Consider the above system. The data pertaining to this is given below →

Generator → $f_{req} = 60$ Hz $\omega_0 = 2\times\pi\times 60$ $X_d = 1.6$ pu

$X_d' = 0.32$ pu $X_q = 1.55$ pu $T_{do}' = 6.0$ s $H = 6$

Network – $X_e = 0.9$ pu [$X_T = 0.1$, $X_{line} = 0.1$ pu]

V (magnitude) = 1.0 E_b (magnitude) = 1 $K_e = 250$

$T_e = 0.025$ sec

loading → $P_m = 0.5$ $P_m = 1.0$

Compute the eigen value of this system for $P_m = 0.5$, 1.0 for the following cases →

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Without AVR ($P_m = 0.5$)

Terminal voltage (θ) angle

$$P_e = P_m = \frac{VE_b \sin\theta}{X} \qquad [P_m = P_e = 0.5]$$

$$0.5 = \frac{1 \times 1}{0.2} \sin\theta \quad \therefore \sin^{-1}(0.1) = 5.73917^\circ$$

Degree to rad $\rightarrow$ $5.73917 \times \frac{\pi}{180^\circ} = 0.100167$

Terminal voltage $V_t \rightarrow$ $V_t = V[\cos\theta + j\sin\theta]$

$V_t = V[\cos 5.73917 + j \sin 5.73917]$ $\quad V = 1$

$V_t = [0.9949 + j\,0.0999]$

$\therefore V_x = 0.9949$ $\quad V_y = 0.0999$

Current phasor (I) $\rightarrow$ $I = \frac{VE_b \angle\theta}{jX_e}$ (while $\angle\theta$)

$$I = \frac{0.9949 + j\,0.0999 - (1 + j0)}{j0.2}$$

$$I = \frac{-0.0051 + j\,0.0999}{j0.2} = \frac{-0.0051}{j0.2} + \frac{j\,0.0999}{j0.2} = 0.4995 + j\,0.0255.$$

$\therefore I_x = 0.4995$, $I_y = 0.0255$.

Current magnitude $= |I| = \sqrt{(I_x)^2 + (I_y)^2}$

$|I| = \sqrt{(0.4995)^2 + (0.0255)^2} = \sqrt{0.250150} = 0.5001$

Current angle $\rightarrow$ $\angle I = \tan^{-1}\left(\frac{I_y}{I_x}\right)$

$\tan^{-1}\left(\frac{0.0255}{0.4995}\right) = \tan^{-1}(0.0510)$

$\angle I = 2.919^\circ$

Rotor angle (δ) $= \angle\delta = \angle V_t + jX_e I$ $\rightarrow$ Apply KVL: $V_t = V_{mc}$

$\delta = \angle (0.9949 + j\,0.0999) + j[1.55(0.4995 + j\,0.0255)]$

$\delta = \angle (0.9949 + j\,0.0999) + j[0.7742 + j\,0.039525]$

$\delta = \angle (0.9949 + j\,0.0999) + 0.7742j - 0.039525$

$\delta = \angle 0.955 + j\,0.8741$

$\delta = \angle 0.955 + j\,0.8741$

$\delta = \tan^{-1}\left(\frac{0.8741}{0.955}\right) = \tan^{-1}(0.91528) = 42.46^\circ$

$= 0.7410$ (in radians)

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→ $I_d = I_x \cos\delta - I_y \cos\delta$

$0.4995 \sin(42.46) - 0.0255 \cos(42.46)$

$0.3372 - (0.73774)\,0.0255$

$0.3372 - 0.01881$

$I_d = 0.31839$ A

$I_q = I_x \cos\delta + I_y \sin\delta$

$0.4995 \cos(42.46) + 0.0255 \sin(42.46)$

$0.4995(0.73774) + 0.01721$

$I_q = 0.3857$ A

$V_d = V_x \sin\delta - V_y \cos\delta$

$0.9949 \sin(42.46) - 0.0999 \cos(42.46)$

$(0.9949)(0.6750) + 0.0999(0.73774)$

$0.67156575 - 0.073700$

$V_d = 0.59786575$ V

$V_q = V_x \cos\delta + V_y \sin\delta$

$0.9949 \cos(42.46) + 0.0999 \sin(42.46)$

$0.9949(0.73774) + 0.0999(0.6750)$

$0.733977 + 0.06743925$

$V_q = 0.80140925$ V

→ $V_q = E_q' - X_d' I_d$

$0.8014095 = E_q' - 0.32 \times 0.31839$

$0.8014095 = E_q' - 0.101884$

$0.8014095 + 0.101884 = E_q'$

$0.90329 = E_q'$

→ C_1, C_2, C_3

→ $C_1 = \dfrac{E_b \sin\delta_0}{X_e + X_d'} = \dfrac{1 \times \sin(42.46)}{0.2 + 0.32} = \dfrac{0.67507}{0.52}$

$C_1 = 1.29802$

→ $C_2 = \dfrac{1}{X_e + X_d'} = \dfrac{1}{0.2 + 0.32} = \dfrac{1}{0.52} = 1.9230$

$C_2 = 1.9230$

→ $C_3 = \dfrac{E_b \cos\delta_0}{X_e + X_q} = \dfrac{1 \cos(42.46)}{0.2 + 1.55} = \dfrac{0.73774}{1.75} = 0.4215$

$C_3 = 0.4215$

→ $K_1 = [(X_q - X_d')\, I_{q0}\, C_1 + (E'_{q0} + (X_q - X_d')\, I_{d0})\, C_3]$

$K_1 = [(1.55 - 0.32)(0.3857)(1.29802) + (0.90329 + (1.55 - 0.32)\,0.31839)\,0.4215]$

$K_1 = [(1.23)(0.5007) + (0.90329 + (1.23)\,0.31839)\,0.4215]$

$K_1 = [0.6158961 + 0.545800]$

$K_1 = 1.16166$

$K_2 = [I_{q0} + (X_q - X_d')\, I_q\, C_2] = [0.3857 + (1.55 - 0.32)\,0.3857 \times 1.9230]$

$[0.3857 + (1.23)(0.74170)]$

$K_2 = 1.297991$

$K_3 = \dfrac{1}{1 + C_2 (X_d - X_d')} = \dfrac{1}{1 + 1.9230(1.6 - 0.32)} = \dfrac{1}{3.46144} = 0.2889$

$K_4 = (X_d - X_d')\, C_1 = -(1.6 - 0.32)\, 1.2982 = 1.66$

$$\begin{bmatrix} \Delta\dot{\delta} \\ \Delta\dot{\omega} \\ \Delta\dot{E}_q' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -43.79 & 0 & -48.93 \\ -0.276 & 0 & -0.57873 \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta\omega \\ \Delta E_q' \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 37.6990 & 0 \\ 0 & 0.166 \end{bmatrix} \begin{bmatrix} \Delta T_m \\ \Delta E_{fd} \end{bmatrix}$$

Eigen Values →

$|sI - A| = 0$

$$sI = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ -43.79 & 0 & -48.93 \\ -0.276 & 0 & -0.57873 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 43.79 & s & 48.93 \\ 0.276 & 0 & s + 0.57 \end{bmatrix}$$

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$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$

$a_{11}M_{11} = \begin{vmatrix} s & 48.93 \\ 0.276\angle 0^\circ & s+0.57870 \end{vmatrix}$

$s^2 + 0.57870s - (48.93)(0.276\angle 0^\circ)$

$s[s^2 + 0.57870s + 13.50468]$

$a_{12}M_{12} = (-1)\begin{vmatrix} 43.79 & 48.93 \\ 0.276 & s+0.57870 \end{vmatrix}$

$-1 * [43.79s + 25.341273 - 13.50468]$

$-43.79s - 11.836593$

$a_{13}M_{13} = 0$

$\Rightarrow s[s^2 + 0.57870s - 13.50468] - [43.79s - 11.836]$

$s^3 + 0.57870s^2 - 13.50468s + 43.79s + 11.836$

$s^3 + 0.57870s^2 + 30.28532s + 11.836$

$s^3 + 0.57870s^2 + 43.79s + 11.836 = 0$

$s_1 = -0.270$

$s_2 = -0.153 + j6.609$ $\quad s_3 = -0.153 - j6.609$ → calculator (stepr Solve)

→ -ve real parts means oscillations decay, ∴ stable system.

without AVR ($P_m = 1$)

→ Terminal voltage angle (θ) →

$P_e = P_m = \frac{VE_b}{X}\sin\theta \qquad [P_m = 1]$

$1 = \frac{1 \times 1}{0.2}\sin\theta$

$\sin^{-1}(0.2) = \theta$

$= 11.5369^\circ$

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→ Terminal voltage $(V_t) = V[\cos\theta + j\sin\theta]$

$1[\cos(11.536) + j\sin(11.5369)]$

$V_t = 0.97979 + j\,0.1999$

$\therefore V_x = 0.97979 \qquad V_y = 0.1999$

→ Current $(I) = \frac{V - E_b}{X_{eq}} = \frac{0.97979 + j0.1999 - 1}{j0.2}$

$= \frac{-0.02021 + j0.1999}{j0.2} = 0.9995 + j0.10105$

$\therefore I_x = 0.9995 \qquad I_y = 0.10105$

→ Rotor angle $(\delta) = \delta \angle V_t + jX_q I$

$\delta = \angle 0.97979 + j0.1999 + j[1.65(0.9995 + j0.10105)]$

$\delta = \angle 0.97979 + j0.1999 + j(1.649225 + j0.1666275)$

$\delta = \angle 0.97979 + j0.1999 + j1.649226 - 0.1666275$

$\delta = \angle 0.8231625 + 1.749125j$

$\therefore \delta = \tan^{-1}\left(\frac{1.749125}{0.8231625}\right)$

$\delta = \tan^{-1}(2.124884) \qquad \delta = 64.79^\circ$

$I_d = I_x\sin\delta - I_y\cos\delta$

$0.9995\sin(64.79) - 0.10105\cos(64.79^\circ)$

$0.9995 \times 0.90475 - 0.10105 \times 0.42593$

$0.904297 - 0.043040 = 0.861257$

$I_d = 0.861257$ pu

$I_q = I_x\cos\delta + I_y\sin\delta$

$0.9995\cos(64.79) + 0.10105\sin(64.79^\circ)$

$0.9995 \times 0.42593 + 0.10105 \times 0.90475$

$0.425717 + 0.091424 = 0.517141$ pu

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$I_q = 0.617141$

$V_d = V_x \sin\delta - V_y \cos\delta$

$0.97979 \sin(64.79°) - 0.1999 \cos(64.79°)$

$0.97979 \times (0.90475) - 0.1999 \times (0.42593)$

$0.886466 - 0.085143$

$V_d = 0.801322$

$V_q = V_x \cos\delta + V_y \sin\delta$

$0.97979 \cos(64.79°) + 0.1999 \sin(64.79°)$

$0.97979 \times (0.42698) + 0.1999 \times (0.90475)$

$0.417321 + 0.180865$

$V_q = 0.598180$

$\rightarrow V_q = E_q' - X_d' I_d$

$0.598180 = E_q' - 0.32 \times 0.861257$

$0.598180 = E_q' - 0.275602$

$0.598180 + 0.275602 = E_q'$

$0.873782 = E_q'$

$\rightarrow C_1 = \frac{E_b \sin\delta_0}{X_e + X_d'} = \frac{1 \times \sin(64.79°)}{0.2 + 0.32} = \frac{0.90475}{0.52} = 1.7399$

$\rightarrow C_2 = \frac{1}{X_e + X_d'} = \frac{1}{0.2 + 0.32} = 1.9230$

$\rightarrow C_3 = \frac{E_b \cos\delta_0}{X_e + X_q} = \frac{1 \times 0.4259}{1.75} = 0.2433$

$\rightarrow K_1 = \left[(X_q - X_d') I_q C_1 + \left(E_q' + (X_q - X_d') I_d \right) C_3 \right]$

$\left[(1.55 - 0.32)(0.617141)(1.73999) + \left(0.873782 + (1.55 - 0.32) 0.861257 \right) 0.2433 \right]$

$\left[(1.23)(0.899820) + (0.873782 + (1.23)\, 0.861257)\, 0.2433 \right]$

$\left[1.1067786 + (0.873782 + 1.059346111)\, 0.2433 \right]$

$\left[1.1067786 + (1.933128)\, 0.2433 \right]$

$K_1 = 1.577169$

$\rightarrow K_2 = \left[I_q + (X_q - X_d') I_q C_2 \right]$

$K_2 = \left[0.617141 + (1.55 - 0.32)(0.617141)(1.9230) \right]$

$K_2 = \left[0.617141 + (1.23)(0.994462) \right]$

$K_2 = \left[0.617141 + 1.223318826 \right]$

$K_2 = 1.740329$

$\rightarrow K_3 = \frac{1}{1 + 1.9230(1.6 - 0.32)} = \frac{1}{1 + 1.9230(1.28)} = \frac{1}{1 + 3.46144} = 0.288889$

$\rightarrow K_4 = (X_d - X_d') C_1 = (1.6 - 0.32)\, 1.7399 = 2.227072$

$$\begin{bmatrix} \Delta\dot{\delta} \\ \Delta\dot{\omega} \\ \Delta\dot{E_q'} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -59.45 & 0 & -66.60 \\ -0.37117 & 0 & -0.5769 \end{bmatrix} \begin{bmatrix} \Delta\delta \\ \Delta\omega \\ \Delta E_q' \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 37.699 & 0 \\ 0 & 0.166 \end{bmatrix} \begin{bmatrix} \Delta T_m \\ \Delta E_{fd} \end{bmatrix}$$

Eigen value $|sI - A| = 0$

$$\begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ -59.45 & 0 & -66.60 \\ -0.37117 & 0 & -0.5769 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 59.45 & s & 66.60 \\ 0.37117 & 0 & s + 0.5769 \end{bmatrix}$$

$|A| \rightarrow a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}$

$a_{11} M_{11} = s \begin{vmatrix} s & 66.60 \\ 0 & s + 0.5769 \end{vmatrix} = s \left[s^2 + 0.5769 s \right]$

$\rightarrow a_{12} M_{12} = -1 \begin{vmatrix} 59.45 & 66.60 \\ 0.37117 & s + 0.5769 \end{vmatrix} = -1 \left[59.45 s + 34.29670 - 24.348 \right]$

$-1 \left[59.45 s + 9.947948 \right]$

$= -59.45 s - 9.947948$

$a_{13} M_{13} = 0$

$s^3 + 0.5769 s^2 + 59.45 s + 9.947948 = 0$

$s_1 = -0.1675$ ✓ $\qquad s_2 = -0.2046 + j\,7.70321$

$s_3 = -0.20468 - j\,7.703321$

$\rightarrow$ -ve real parts means oscillation decay, $\therefore$ stable system

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With AVR $P_m = 0.5$

$K_e = 250$ $T_e = 0.025$ s

$$K_5 = \left[\frac{V_d}{V} X_q C_3 - \frac{V_q}{V} C_1 X_d'\right]$$

$$K_5 = \frac{0.5978575}{1}(1.65)(0.4215) - \frac{0.8014095}{1}(1.2982)(0.32)$$

$$K_5 = 0.390595251 - 0.332292474$$

$$K_5 = 0.059670$$

$$K_6 = \left[\frac{V_{q0}}{V_0}(1 - C_2 X_d')\right] = \left[\frac{0.8014095}{1}(1 - (1.9230)0.32)\right]$$

$$= [0.8014095(1 - 0.61536)]$$

$$K_6 = 0.308254116$$

$$\begin{bmatrix}\Delta\dot{\delta}\\ \Delta\dot{\omega}\\ \Delta\dot{E}_q'\\ \Delta\dot{E}_{fd}\end{bmatrix} = \begin{bmatrix}0 & 1 & 0 & 0\\ -43.79 & 0 & -48.93 & 0\\ -0.276 & 0 & -0.57870 & 0.1666\\ -576.7 & 0 & -3082.54 & -40\end{bmatrix}\begin{bmatrix}\Delta\delta\\ \Delta\omega\\ \Delta E_q\\ \Delta E_{fd}\end{bmatrix} + \begin{bmatrix}0 & 0\\ 31.693 & 0\\ 0 & 0\\ 0 & 16{,}000\end{bmatrix}\begin{bmatrix}\Delta T_m\\ \Delta V_{ref}\end{bmatrix}$$

eigen values $= |sI - A|$

$$\begin{bmatrix}s & 0 & 0 & 0\\ 0 & s & 0 & 0\\ 0 & 0 & s & 0\\ 0 & 0 & 0 & s\end{bmatrix} - \begin{bmatrix}0 & 1 & 0 & 0\\ -43.79 & 0 & -48.93 & 0\\ -0.276 & 0 & -0.57870 & 0.1666\\ -576.7 & 0 & -3082.54 & -40\end{bmatrix} = \begin{bmatrix}s & -1 & 0 & 0\\ 43.79 & s & 48.93 & 0\\ 0.276 & 0 & s+0.57870 & -0.1666\\ 576.7 & 0 & 3082.54 & s+40\end{bmatrix}$$

$a_{11}M_{11} = \begin{vmatrix}s & 48.93 & 0\\ 0 & s+0.57870 & -0.1666\\ 0 & 3082.54 & s+40\end{vmatrix} = s\left[s(s+0.57870)(s+40) - (-0.1666)(3082.54)\right]$

$$= s\left[s^3 + 40.57870s^2 + 536.748\right] \Rightarrow s\left[s^3 + 40.57870s^2 + \frac{699}{699}536.748s\right]$$

$$= s^4 + 40.57870s^3 + 536.748s^2$$

$\rightarrow a_{11}M_{11} - a_{12}M_{12}$

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$a_{12}M_{12} \Rightarrow -1\begin{bmatrix}43.79 & 48.93 & 0\\ 0.276 & s+0.57870 & -0.1666\\ 576.7 & 3082.54 & s+40\end{bmatrix}$

$$\Rightarrow -\left[(43.79)\begin{vmatrix}s+0.57870 & -0.1666\\ 3082.54 & s+40\end{vmatrix}\right] = -43.79\left[\frac{(s+0.57870)(s+40) - (3082.54)}{(-0.1666)}\right]$$

$$\Rightarrow 43.79\left[s^2 + 40s + 0.57870s + 23.148 + 513.5511\right] = 43.79\left[s^2 + 40.57870s + 536.6991\right]$$

$$= 43.79s^2 + 1776.941s + 23502.053$$

$$48.93\begin{bmatrix}0.276 & -0.1666\\ 576.7 & s+40\end{bmatrix} = 48.93\left[0.276s + 11.04 + 96.07822\right]$$

$$= 48.93\left[0.276s + 107.11822\right] = \left[13.50468s + 5241.294\right]$$

$M_{12} = M_{11}(\text{from } a_{11}M_{11}) - M_{12}(\text{from } a_{12}M_{12}) = 43.79s^2 + 1776.941s + 23502.053 - 13.50468s - 5241.294$

$$\therefore 43.79s^2 + 1776.941s + 23502.053 - 13.50468s - 5241.294$$

$$43.79s^2 + 1763.43632s + 18260.759$$

$|A| = sM_{11} - (-1)M_{12}$

$$|A| = s^4 + 40.57870s^3 + 536.748s^2 + 1\left[43.79s^2 + 1763.43632s + 18260.759\right]$$

$$s^4 + 40.57870s^3 + 536.748s^2 + 43.79s^2 + 1763.43632s + 18260.759 = 0$$

$$s^4 + 40.57870s^3 + 580.53832s^2 = 0$$

$$s^4 + 40.57870s^3 + 580.489s^2 + 1763.43632s + 18260.759 = 0$$

$$\therefore s_1 = -19.94 + j10.9520 \qquad s_3 = -0.3443 + j5.9288$$

$$s_2 = -19.94 - j10.9520 \qquad s_3 = -0.3443 - j5.9288$$

Eigen values

$\therefore$ all 4 eigenvalues have -ve real parts $\therefore$ system is stable.

With AVR $P_m = 1$

$K_e = 250$ $T_e = 0.025$

$$K_5 = \left[\frac{V_d}{V} X_q C_3 - \frac{V_q}{V} C_1 X_d'\right] = \left[0.801322(1.55)(0.2432) - 0.598180(1.739)(0.32)\right]$$

$$K_5 = 0.308190 - 0.332875205 = -0.024685$$

$$K_6 = \left[\frac{V_{q0}}{V_0}(1 - C_2 X_d')\right] = \left[0.598180(1 - (1.9230)(0.32)\right]$$

$$0.598180(0.38464)$$

$$K_6 = 0.2300839$$

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A matrix →

$$\begin{bmatrix} s & -1 & 0 & 0 \\ 59.45 & s & 65.60 & 0 \\ 0.37117 & 0 & s+0.5769 & -0.1666 \\ -306.852 & 0 & -2300.839 & s+40 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} s & 65.60 & 0 \\ 0 & s+0.5769 & -0.166 \\ 0 & 2300.839 & s+40 \end{bmatrix}$$

$$s\left[(s+0.5769)(s+40) - (-0.166)(2300.839)\right]$$

$$s\left[s^2+40s+0.5769s+23.076+383.319774\right]$$

$$s\left[s^2+40.5769s+406.395774\right] = s^3+40.5769s^2+406.395774s$$

$$M_{12} = \begin{bmatrix} 59.45 & 65.60 & 0 \\ 0.37117 & s+0.5769 & -0.166 \\ -306.852 & 2300.839 & s+40 \end{bmatrix}$$

$$59.45\left[(s+0.5769)(s+40) - (2300.839)(-0.1666)\right]$$

$$59.45\left[s^2+40s+0.5769s+(0.5769)(40)+383.319774\right]$$

$$59.45\left[s^2+40.5769s+23.076+383.319774\right]$$

$$59.45\left[s^2+40.5769s+406.395774\right]$$

$$59.45s^2+2412.296s+24160.22897$$

$$\Rightarrow 65.60\left[(0.37117)(s+40) - (-306.852)(-0.1666)\right]$$

$$65.60\left[0.37117s+14.8468-51.121543\right]$$

$$65.60\left[0.37117s-36.274743\right]$$

$$24.34908s-2379.623154$$

$$\therefore M_{12} = \left[59.45s^2+2412.296s+24160.22897-24.34908s+2379.623154\right]$$

$$= \left[59.45s^2+2387.94692s+26539.8521\right]$$

$$|A| = s|M_{11}| - (-1)|M_{12}|$$

$$s^4+40.5769s^3+406.395774s^2+59.45s^2+2387.94692s+26539.8521$$

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Classical Model $P_m = 0.5$

$\rightarrow \theta \Rightarrow \quad P_m = P_e = \frac{V \times E_b'}{X_{eq}} \sin\theta \quad = 0.5 = \frac{1 \times 1}{0.2} \sin\theta$

$0.1 = \sin\theta \quad ; \quad \sin^{-1}(0.1) = \theta \quad \rightarrow \theta = 5.739178^\circ$

$\rightarrow V(t) = 1 \angle 5.739178^\circ \Rightarrow \bar{V} = 1(\cos\theta + j\sin\theta) \rightarrow \bar{V} = 0.9949 + j0.0999$

$\rightarrow I = \frac{V - E_b}{X_e} = \frac{0.9949 + j0.0999 - 1}{j0.2} = 0.4995 + j0.0255$

Apply KVL $\rightarrow E = V + jX_{eq}I \quad \ldots (q_m)$.

$E = 0.9949 + j0.0999 + j(0.32)(0.4995 + j0.0255)$

$E = 0.9949 + j0.0999 + j[0.15984 + j0.00816]$

$E = 0.9949 + j0.0999 + j0.159849 - 0.00816$

~~$E = 0.9949$~~ $E = 0.98674 + j0.259749$

$\text{Mag}[E'] = \sqrt{(0.98674)^2 + (0.259749)^2}$

$= \sqrt{0.973655827 + 0.067469543}$

$= \sqrt{1.04112537} = 1.0203555 \text{ V}$

$\delta = \tan^{-1}\left[\frac{0.259749}{0.98674}\right] = \tan^{-1}[0.263239556] = 14.747^\circ$

$\rightarrow K_s = \frac{E_b E_b' \cos\delta}{(X_1 + X_2)\, X_{eq}} = \frac{1.0203555 \times 1 \cos(14.747)^\circ}{0.52}$

$K_s = 1.962222115 \times 0.967059$

$K_s = 1.8975$

$S - \left(\frac{\pi f_0 K_s}{H}\right)$

$|sI - A| = s^2 - \left(\frac{-\pi f_0 K_s}{H}\right) = 0 \Rightarrow s^2 + 71.53406472 = 0$

$s = \sqrt{-71.53406472}$

$s = \pm j\, 8.45778.$

$\frac{\pi f_0 K_s}{H}$

$= 0 + j\,8.45778 \quad , \quad 0 - j\,8.45778.$

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Classical Model $P_m = 1$

→ $\Theta \Rightarrow P_m = P_e = \frac{V E_0 \sin\Theta}{X_{eq}} \Rightarrow 1 \times 0.2 = \sin\Theta$

$\Theta = 11.536$

$V \Rightarrow 0.97987 + j0.1999$ (previously calculated)

$I \Rightarrow 0.9995 + j0.10105$ (already calculated).

→ $E = V + jX_d' I$

$E' = 0.97979 + j0.1999 + j\left[(0.32)(0.9995 + j0.10105)\right]$

$E' = 0.97979 + j0.1999 + j\left[0.31984 + j\,0.032336\right]$

$E' = 0.97979 + j0.1999 + j0.31984 - 0.032336$

$E' = 0.947454 + j\,0.51974$

$E'(mag) = \sqrt{(0.947454)^2 + (0.51974)^2}$

$\sqrt{0.897669082 + 0.270129667}$

$\sqrt{1.167799749} = 1.0806473.$

$\therefore K_s = \frac{E' V \sin\delta_0}{X_{eq}} = \frac{1.0806473 \cos(28.474)}{0.52} = 1.826779.$

(*) $\delta = \tan^{-1}\left(\frac{0.51974}{0.947454}\right) = \tan^{-1}(0.54856489) \simeq 28.474$

$|sI - A| = s^2\left(-\frac{\pi f^0 K_s}{H}\right) = 0 \Rightarrow s^2 + \frac{344.3397292}{5} = 0$

$s^2 = -344.3397292$ 68.86794583

$s = \pm j\ 8.29867.$

$\therefore s = 0 + j\,8.29867$

$s = 0 - j\,8.29867.$

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→ Damping and Synchronizing torque analysis.

In the classical model, one model is considered - a single machine connected to infinite bus. We will not include damper winding here.

But when we get eigenvalues like $\lambda = -0.16 \pm j6.28$ instead of $\lambda = \pm j6.28$. Here damping is we get one real part that means damping is created. Damping is created by field winding. Rotor oscillates, the field flux also changes, because of flux dynamics it cannot change instantaneously. This lag in the field flux creates a torque that opposes rotor oscillations. That is called electromagnetic damping.

→ Swing mode - when you get eigen values $\lambda_{1,2} = \sigma \pm j\omega$, this pair corresponds to rotor oscillation. Rotor swings around its equilibrium position. It's called swing mode.

→ Synchronizing τ → when rotor angle changes $\Delta\delta$, electrical τ changes. Part of that τ tries to pull rotor back towards synchronism. This is called synchronizing τ. [$\Delta T_s = K_s \Delta\delta$] It acts like a spring - when rotor speed changes $\Delta\omega$, another τ is produced opposing speed deviation. This is called damping τ.

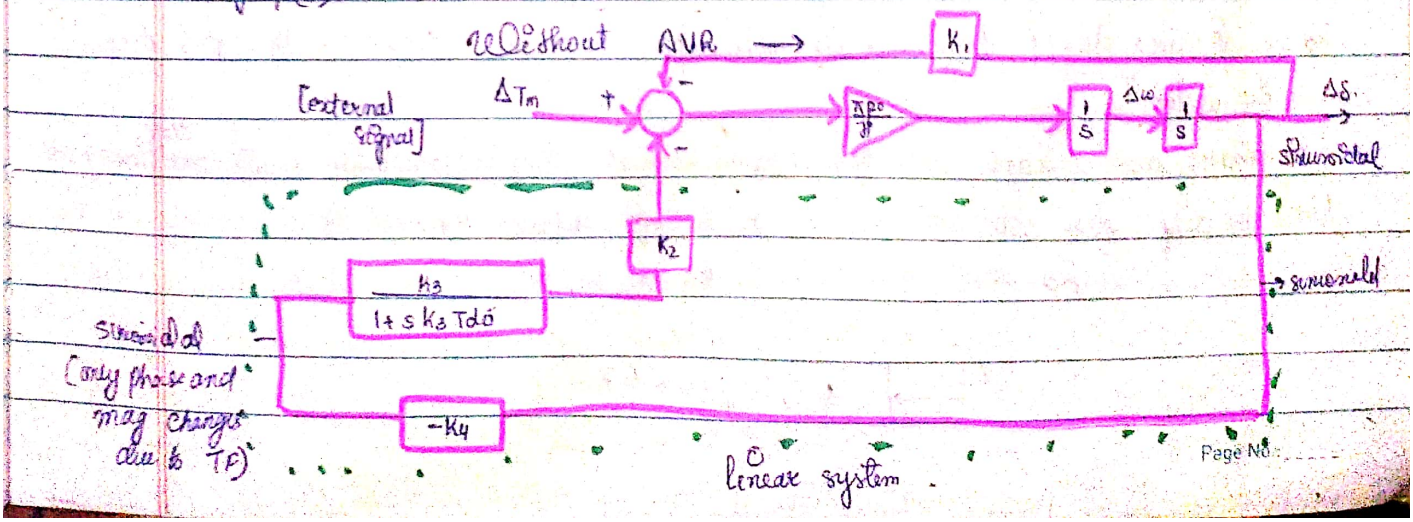
$$\boxed{\Delta T_e = K_s \Delta\delta + K_d \Delta\omega}$$

$\lambda_{1,2} = \pm\sigma \pm j\omega$ → the real part (σ) controls damping, Imaginary part (ω) controls oscillation frequency. Rotor angle $\Delta\delta$ = physical meaning - ω controls how fast oscillations occurs - large ω - many oscillation, small ω - slow oscillation

Oscillation freq / Swing freq → After a disturbance, the rotor angle δ does not immediately settle. It swings $\delta_0 \to \delta_1 \to \delta_2 \to \delta_0$. This back- and forth motion has its own freq that called oscillation freq or swing freq.

→ Mathematically → [stiff rotor is damping when no damping winding]

Damping and synchronizing torque analysis developed by Demello and Concordia, is a freq domain method used to explain the effect of excitation system on the oscillatory stability of SMIB. Basic Idea → Consider the generating operating at equilibrium. Following a small disturbance the rotor angle oscillates represents by (ω). Assume that the rotor angle oscillates at freq (ω).



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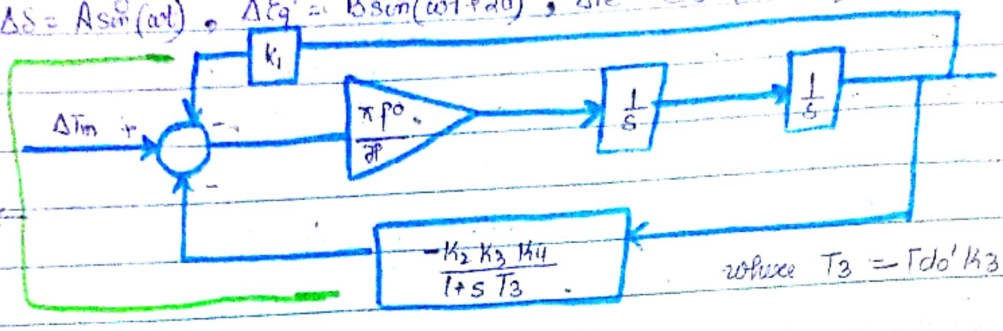
we have $\lambda = -\sigma \pm j\omega$. Then the solution of the system contains $e^{(-\sigma - j\omega)t}$ and $e^{(-\sigma + j\omega)t}$ combining them gives $e^{-\sigma t}\sin(\omega t + \phi)$ or $e^{-\sigma t}\cos(\omega t + \phi)$

So $\Delta\delta(t)$ is an oscillation (a damped sine wave)

→ Why are these signals also sinusoidal? As whole system is linearised. For a linear sys. we are analysing the oscillatory response (swing mode) of the system after a disturbance. Then every other variable also oscillates at the same freq only amplitude changes, phase changes.

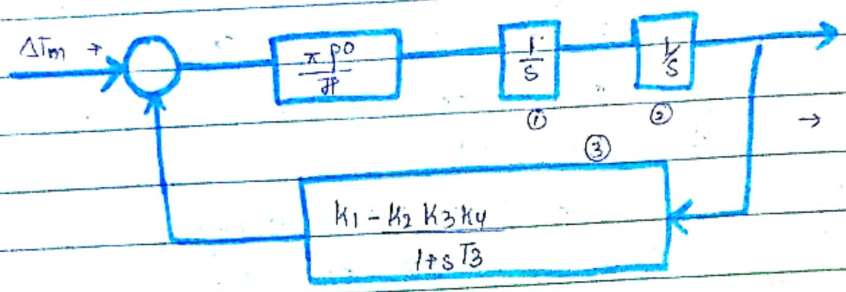
For eg - $\Delta\delta = A\sin(\omega t)$, $\Delta E_q' = B\sin(\omega t + 20)$, $\Delta T_e = C\sin(\omega t - 165^\circ)$



They are in parallel

where $T_3 = T_{do}' K_3$

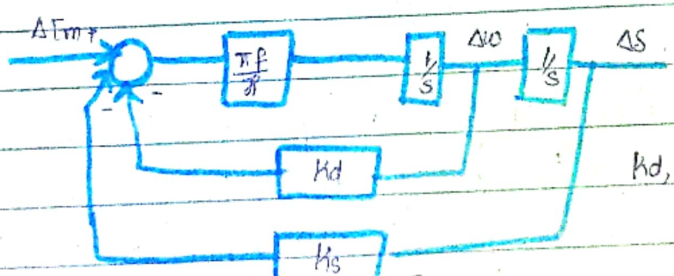
Damping axis



→ This is 3 order system.

Damping and synchronizing τ analysis is based on approximate second order of the system. (AVR or noAVR - 2nd order system) [So we convert 3 order to 2 order].

Put $\frac{K_1 - K_2 K_3 K_4}{1+sT_3} = G_1(s)$



K_d, K_s — are constant depend on freq

Hence $\Delta T_m = K_d \Delta\omega + K_s \Delta\delta$ [$K_s \Delta\delta$ is called synchronizing τ, if rotor angle increases $\Delta\delta > 0$ then electrical torque changes and tries to pull the rotor back. Like a spring $K_d \Delta\omega$ is called damping torque. If rotor speed increases $\Delta\omega > 0$, then this torque opposes the motion].

K_d and K_s are constants and they depend on the freq of oscillation. For one coeff load have diff freq oscillation ∴ we have diff K_d & K_s for K_d and K_s

$K_d \rightarrow K_d(s)$ $\quad K_s \rightarrow K_s(s)$.

$\Delta\delta \rightarrow$ [$G_1(s)$] $\rightarrow \Delta T_e$

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$G_1(s) = \frac{\Delta T_e}{\Delta \delta}$

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$\Rightarrow \Delta T_e = G_1(s)\,\Delta\delta$ [so $G_1(s)$ tells us if rotor angle changes, how much electrical τ changes]

$G_1(s) = K_s + j K_d(\omega)$

$\therefore \Delta T_e = [K_s + j K_d(\omega)]\,\Delta\delta \rightarrow \Delta T_e = K_s \Delta\delta + j\omega K_d \Delta\delta \longrightarrow (1)$

$\Delta\omega \rightarrow \boxed{\frac{1}{j\omega}} \rightarrow \Delta\delta$

$\frac{\Delta\delta}{\Delta\omega} = \frac{1}{j\omega}$

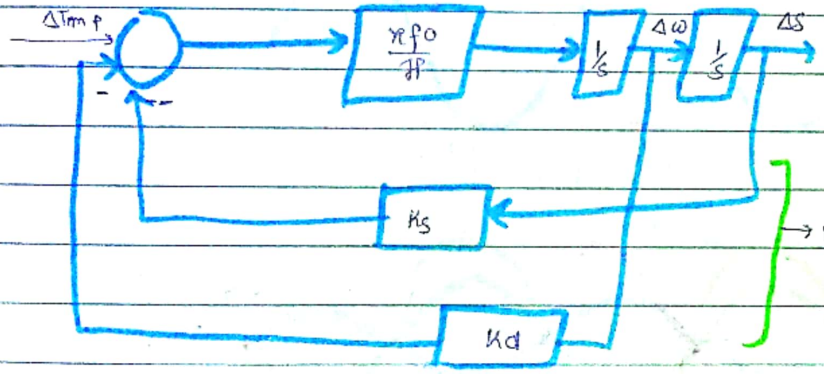
$\Delta\omega = j\omega\,\Delta\delta \rightarrow$ put above. (1)

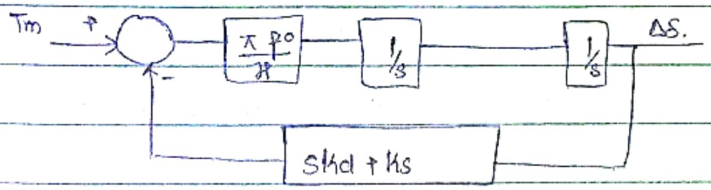
$\Delta T_e = K_s \Delta\delta + j\Delta\omega K_d \Delta\delta$

$\Delta T_e = K_s \Delta\delta + \Delta\omega K_d$

→ How to get K_d, K_s from $G_1(s)$? K_s is the real part of $G_1(s)$, → while as K_d is imag

$K_s = \text{Re}\left[G_1(s)\right]_{s=j\omega}$ $\quad K_d = \frac{1}{\text{omega}(\omega)}\,\text{Img}\left[(G(s))\right]_{s=j\omega}$





∴ Characteristic eq →

$s^2 + \frac{2f_o K_d}{H} s + \frac{2f_o}{H} K_s = 0$

Keep in mind K_s and K_d are not constants they are fns of rotor oscillation freq (ω) as TF $G_1(s)$ contains terms $(1+sT_d K_d)$ and $(1+sT_e)$.

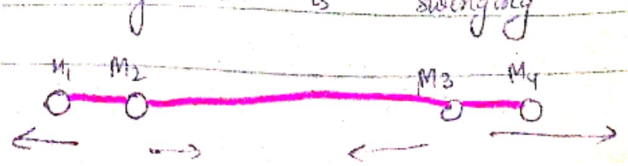
— o — o —

Mode

Suppose a disturbance occurs. Generators do not instantly settle. They swing δ's. The pattern in which generators swing is called a mode of oscillation

① Local mode – generator ⟷ stiff grid. One generator oscillates against the rest of the system. Only one generator swinging

② Inter area mode.



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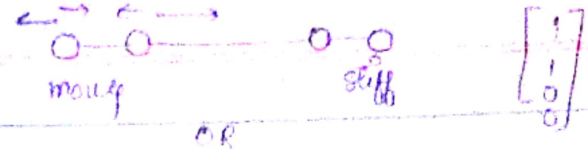
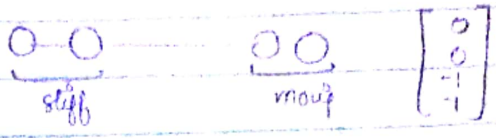
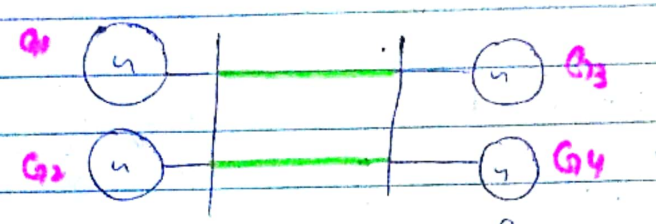
Intraplant mode

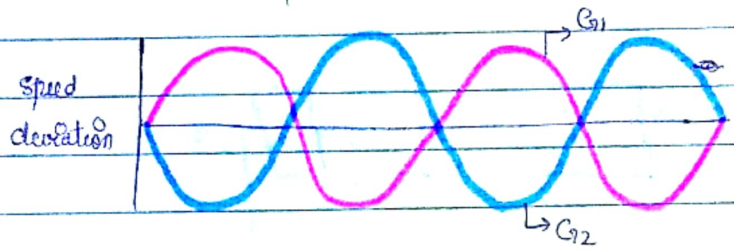


or

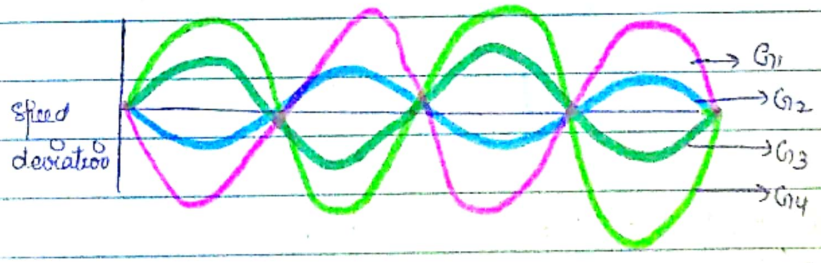


Swing modes are natural oscillation patterns describing how the rotors of the system move relative to one another. Each swing mode is characterized by a frequency that specifies the pattern of relative rotor motion.





Interarea →



n- machine system there are n-1 swing mode. Common mode is not a swing mode

A generator connected to a bus has the following constant at particular frequency → $K_1 = 1.8$, $K_2 = 1.5$, $K_3 = 0.8$, $K_4 = 1.2$, $T_{do'} = 5s$. The generator oscillates at $\omega = 6$ rad/s. Determine the synchronizing coefficient K_s and damping coefficient K_d.

Sol → SMIB AVR system →

$$G(s) = \frac{K_1 - K_2 K_3 K_4}{1 + s K_3 T_{do'}}$$

Now calculating K_s and K_d. Evaluate $G(j\omega) = \left[\frac{\quad}{s = j\omega}\right]$

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So $G(j\omega) = K_1 - \frac{K_2 K_3 K_4}{1 + j\omega K_3 T_{do'}}$

$$G_1(j\omega) = \frac{1.8 - 1.44}{1 + j(6)(0.8)(5)}$$

$$G_1(j\omega) = \frac{1.8 - 1.44}{1 + j24}$$

$$G_1(j\omega) = 1/8 \;/\; 1/44$$

$$G_1(j\omega) = \frac{1.8(1 + j24) - 1.44}{1 + j24}$$

$$G_1(j\omega) = \frac{1.8 + j43.2 - 1.44}{1 + j24} = \frac{0.36 + j43.2}{1 + j24}$$

$$\frac{0.36 + j43.2 \times (1 - j24)}{1 + j24 \times (1 - j24)} = \frac{1037.16 + j34.56}{577}$$

$$= 1.79750 + j\,0.05989$$

$\therefore G(j\omega) = K_s + j K_d \;\Omega$

$K_s = 1.79750$

$K_d = 0.009966 \approx 0.01$ (approx)

$jG_1(j\omega) = j K_d \;\Omega$

$\frac{0.05989}{6\,\Omega} = K_d$

Assignment →

$K_1 = 1.4 \quad K_2 = 2.0 \quad K_3 = 0.7 \quad K_4 = 1.5 \quad T_{do'} = 6s \quad \Omega = 4$ rad/s

$$G_1(j\Omega) = K_1 - \frac{K_2 K_3 K_4}{1 + j\Omega K_3 T_{do'}} = 1.4 - \frac{(2)(0.7)(1.5)}{1 + j(4)(0.7)(6)}$$

$$G_1(j\Omega) = 1.4 - \frac{2.1}{1 + j16.8} = \frac{1.4(1 + j16.8) - 2.1}{1 + j16.8}$$

$$G(j\Omega) = \frac{1.4 + j23.52 - 2.1}{1 + j16.8} = \frac{-0.7 + j23.52 \times 1 - j16.8}{1 + j16.8 \times 1 - j16.8} =$$

$$G(j\Omega) = \frac{394.436 + j35.28}{203.24} = 1.3925 + j\,0.12455$$

$G(j\Omega) = 1.3925 + j\,0.12455$

$K_s = 1.3925$

$K_d = \frac{0.12455}{4} = 0.0311$

Assignment

Soln → 400kV AVR

$K_1 = 1.5$, $K_2 = 1.2$, $K_3 = 0.8$, $K_4 = 1.0$, $K_5 = -0.3$, $K_6 = 5$

$T_{do'} = 6s$, $K_E = 50$, $T_E = 0.05s$, $\omega = 5 \text{ rad/s}$

$$\rightarrow G(s) = \frac{K_1 - K_2 K_3 \left(K_4 + \frac{K_6 K_5}{1+sT_E} \right)}{1 + s K_3 T_{do'} + \frac{K_3 K_6 K_E}{1+sT_E}}$$

$$G(j\omega) = \frac{1.5 - (1.2)(0.8)\left[1 + \frac{(-0.3)(50)}{1+j(5)(0.05)}\right]}{1 + j(5)(0.8)(6) + \frac{(0.8)(0.5)(50)}{1+j(5)(0.05}}$$

$$\Rightarrow \frac{1.5 - 0.96\left[1 - 15\right]}{1 + j24 + \frac{20}{1+j0.25}} \Rightarrow \frac{1.5 - \left[\frac{-13.44}{1+j0.25}\right]}{\frac{1+j24+20}{1+j0.25}}$$

Solving numerator → $1.5 - \left[\frac{-13.44}{1+j0.25}\right]$

$$\Rightarrow 1.5 - \left[\frac{-13.44 \times 1 - j0.25}{1+j0.25 \times 1-j0.25} = \frac{-13.44 + j3.36}{1.0625}\right]$$

$$1.5 - \left[-12.64941 + j3.16235\right] \longrightarrow (1)$$

Solving denominator → $1 + j24 + \frac{20}{1+j0.25} \times \frac{1-j0.25}{1-j0.25}$

$$1 + j24 + \left[\frac{20 - j5}{1.0625}\right]$$

$$1 + j24 + 18.8235 - j4.70588$$

$\Rightarrow$

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$19.8235 + j\,19.294 \longrightarrow (2)$

eq (1) ÷ (2)

$$\rightarrow 1.5 - \left[\frac{[-12.64941 + j\,3.16833] \times 19.8235 - j\,19.294}{19.8235 + j\,19.294 \times 19.8235 - j\,19.294}\right] =$$

$$1.5 - \left[\frac{-189.741 + j\,306.746}{765.2295}\right]$$

$$1.5 - [-0.24795 + j\,0.40085]$$

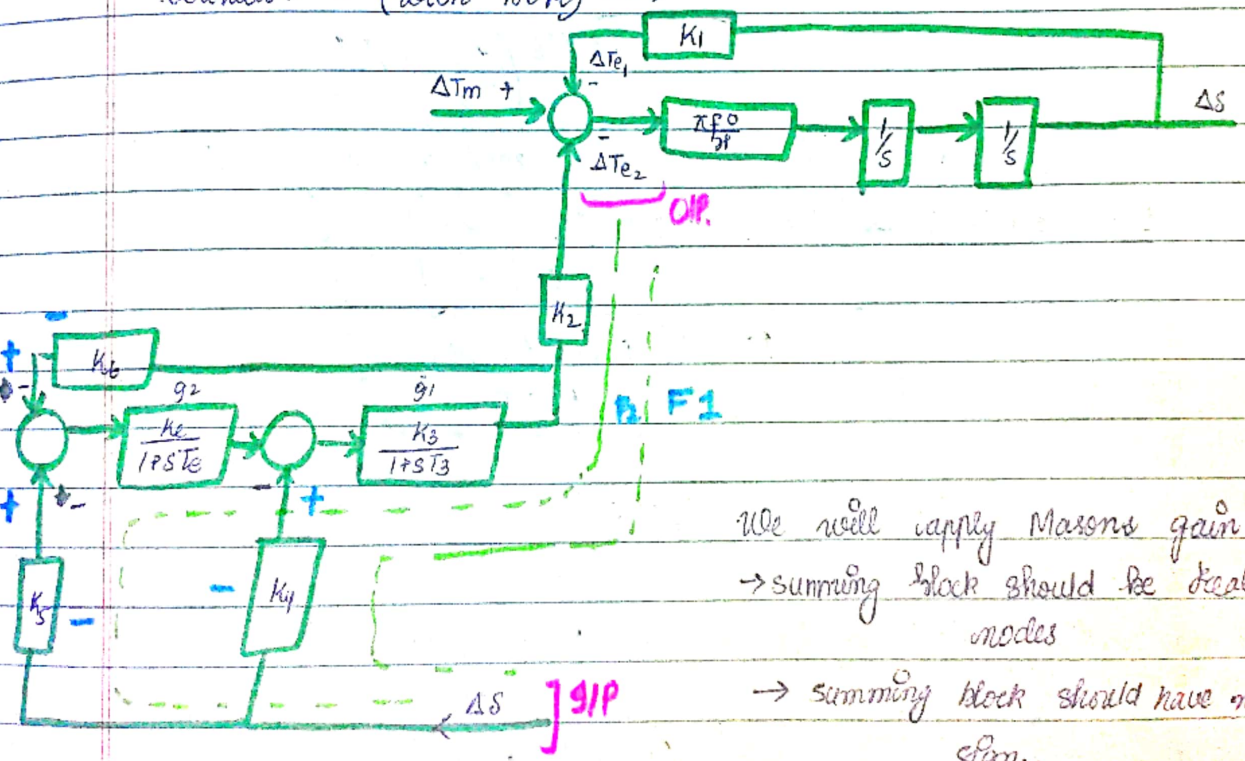
$$1.5 + 0.24795 + j\,0.40085$$

$$K_{sc} = 1.74 + j\,0.4008$$

$$K_s = 1.74$$

$$K_d = \frac{0.4008}{5} = 0.08016$$

Derivation → (with DVR) →



We will apply Masons gain
→ summing block should be treated as nodes
→ summing block should have no -ve sign.

F1 → forward path 1 F2 - forward path 2.

$$\Delta T_e = \underbrace{K_1 \Delta S}_{\Delta T_{e1}} + \underbrace{K_2 \Delta E_q'}_{\Delta T_{e2}} \longrightarrow (1)$$

$$\Delta T_{e1} = K_1 \Delta S \longrightarrow (2)$$

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Applying masons gain formula path 1

$$\frac{\Delta Te_2}{\Delta S} = \frac{-\left[K_4 K_2 G_1\right]\Delta_1 - \left[K_2 K_5 K_3 G_1 G_2\right]\Delta_2}{\Delta}$$

$$\frac{\Delta Te_2}{\Delta S} = \frac{-\left[K_2 K_4 G_1\right]\Delta_1 - \left[K_2 K_5 G_1 G_2\right]\Delta_2}{1-(-K_6 G_1 G_2)}$$

$$\Delta_1 = \Delta_2 = 1$$

(Taking Common) $$\frac{\Delta Te_2}{\Delta S} = \frac{-K_2 G_1\left[K_4 + K_5\left(\frac{K_e}{1+sT_e}\right)\right]}{1+K_6 G_1 G_2}$$

$$\frac{\Delta Te_2}{\Delta S} = \frac{-K_2\left[\frac{K_3}{1+sT_3}\right]\left[K_4 + K_5\left(\frac{K_e}{1+sT_e}\right)\right]}{1+K_6\left(\frac{K_3}{1+sT_3}\right)\left(\frac{K_e}{1+sT_e}\right)}$$

multiply numerator and denominator by $(1+sT_3)$

$$\frac{\Delta Te_2}{\Delta S} = G'(s) = \frac{-K_2 K_3\left[K_4 + \frac{K_5 K_6 K_e}{1+sT_e}\right]}{(1+sT_3) + \frac{K_3 K_6 K_e}{1+sT_e}}$$

$$G_1(s) = K_1 + G_1'(s)$$

$$G_1(s) = K_1 - \frac{K_2 K_3\left[K_4 + \frac{K_5 K_6 K_e}{1+sT_e}\right]}{(1+sT_3) + \frac{K_3 K_6 K_e}{1+sT_e}}$$

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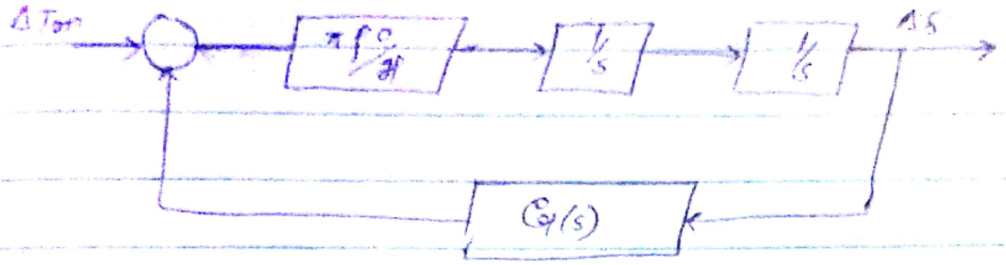

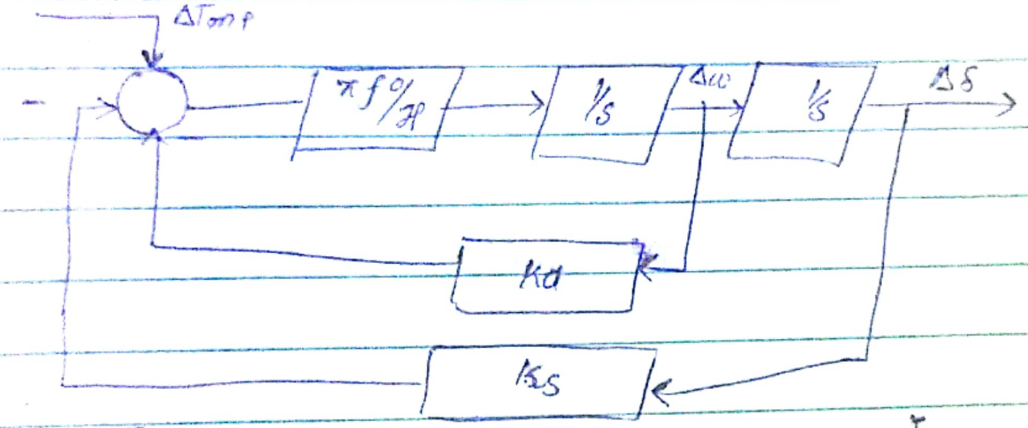

$$K_s = \text{Real}\left[C_q(s)\right]_{s=j\omega} \quad ; \quad K_d = \frac{1}{\omega}\,\text{Im}\left[C_q(s)\Big|_{s=j\omega}\right] \quad (+)$$